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This Scientific article is based on the steps by steps process for the identification of the different features on the Cameroon map with the use of Quantum Geographical Information System(QGIS)

On this article we have the steps to:

- Identify the 10 regions on the Cameroon map including it's water bodies,coastal lines,Roads,different locations,Divisions and Subdivisions.
- Measure the area of a particular Zone on the map
- Calculate the distance between two points on a map
- Calculate the population density of the 10 regions of Cameroon and represent it on the map

I. INTRODUCTION

a. What is GIS?

GIS fully known as Geographical Information System is a computer system that analyses and displays geographically referenced informations.

The field of Geographic Information Systems (GIS) started in the 1960s as computers and early concepts of quantitative and computational geography emerged. Geographic information systems have emerged in the last decade as an essential tool for urban and resource planning and management. The first computerized GIS in the world was developed by **Roger Tomlinson in 1963**. Their capacity to store, retrieve, analyse, model and map large areas with huge volumes of spatial data has led to an extraordinary proliferation of applications. Geographic information systems are now used for “land use planning, utilities management, ecosystems modelling, landscape assessment and planning, transportation and infrastructure planning, market analysis, visual impact analysis, facilities management, tax assessment, real estate analysis and many other applications”. Functions of GIS include: “data entry, data display, data management, information retrieval and analysis”. An example of a software enabling these function above is **QGIS**.

b. What is QGIS ?

QGIS fully known as Quantum Geographical information system is a **free and open-source geographic information system (GIS)** that allows you to create, edit, visualize, analyze, and publish geospatial information on Windows, macOS, Linux, BSD, and mobile devices . It is a powerful tool for working with maps and spatial data, and it is used by a wide range of professionals, including geographers, cartographers, environmental scientists, and urban planners. QGIS is designed to be user-friendly and easy to learn, with a wide range of features and plugins that can be customized to suit your needs .

The project was born in May 2002 and was established as a project on SourceForge in June the same year. QGIS is available to anyone with access to a personal computer. QGIS currently runs on most Unix platforms, Windows, and macOS.

QGIS is developed using the Qt toolkit and C++. This means that QGIS feels snappy and has a pleasing, easy-to-use graphical user interface (GUI).

II. METHODOLOGY:

Methodology refers to the systematic and theoretical analysis of the methods applied to a field of study. It is a detailed description of the methods used to conduct research, including the theoretical framework, research design, data collection, and data analysis.

The methodology used for the distribution of work amongst the members of this team is **SCRUM**.

What is SCRUM? **Scrum** is a methodology that is used in agile software development. It is a framework that helps teams work together to achieve a common goals.

The following steps were followed according to the scrum methodology for the distribution of work amongst our group members:

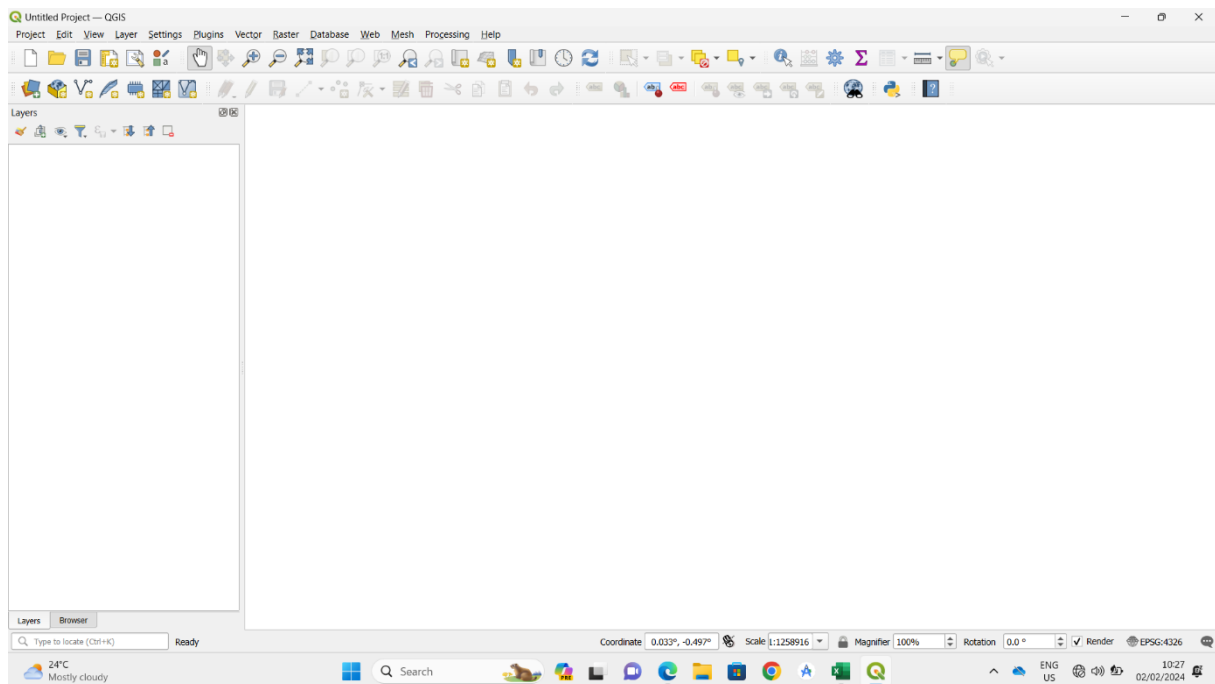
1. **Creation of a product backlog** : Here we came out with the list of the different tasks that were to be achieved according to what our team was all about (QGIS). This included the creation of a Cameroon map and identifying the following features on it;
 - Population density
 - All the roads and rail ways in Cameroon
 - The different coastal lines
 - The 10 regions, their subdivisions and departements
 - Water Bodies in Cameroon
 - Preparing a scientific Article
 - Calculation of the distance between lines, points and particular areas.
 - Calculating the surface areas of particular Zones.
 - Identifying features
2. **Planning of the sprint**: Here, according to the different tasks stated in our product backlog we decided to organise ourselves on how we were going to proceed to the execution of the product backlog. We decided to follow the following steps respectively:
 - Discuss about all our different ideas concerning the theme
 - Carry out research on what the theme is all about
 - Download the required software (QGIS 3.34.3)
 - Learn how to use this software to execute our product backlog
 - Download the necessary files in order to carry out the different tasks (Here we had to download the shp.files(shape files) for Cameroon specifically for its roads, water bodies, Coastal lines, railways, Regions, Divisions, Subdivisions and departements.
 - Start performing the different task stated in the product backlog
3. **Distribution of work**: There were 9 members in the group let's just identify these nine members as **A,B,C,D,E,F,G,H,I** respectively. The work was shared as follows:

❖ A = Downloaded the necessary shape files

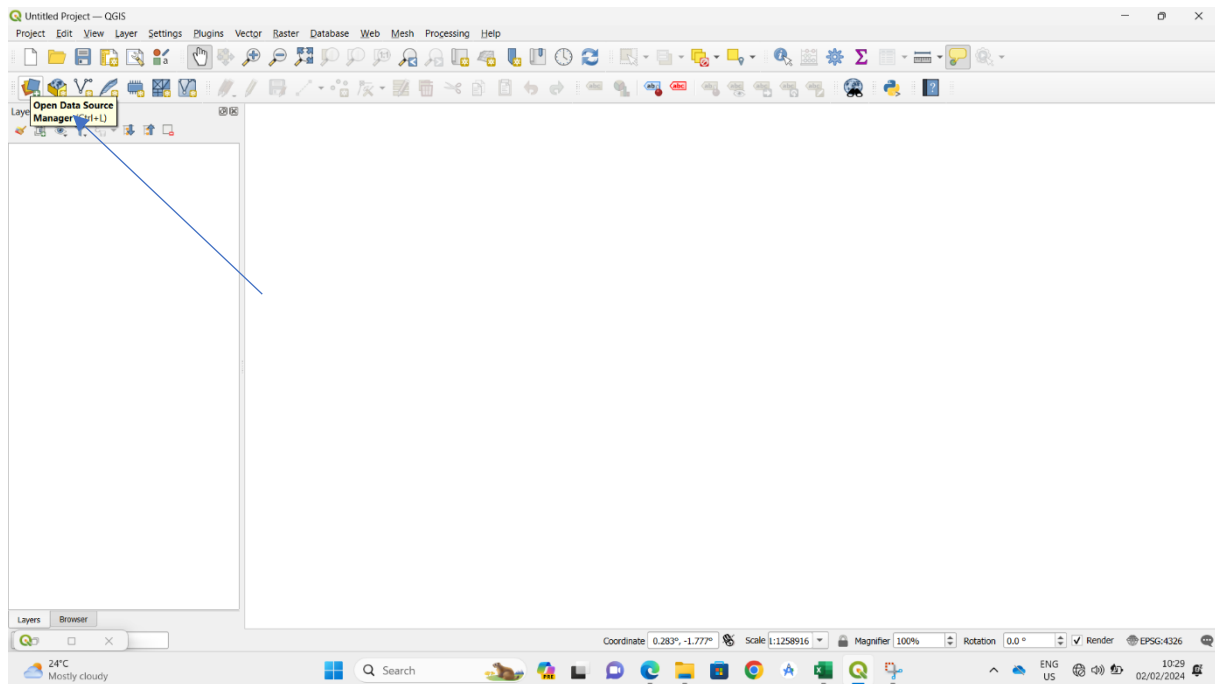
- ❖ B = Collected the data for the different populations areas
- ❖ C = Carried out the execution of the different tasks on QGIS
- ❖ D = Prepared the Sceintific Article
- ❖ E = Identified the different regions on the map(different colours)
- ❖ F = Identified the divisions, subdivisions and departements of this regions.
- ❖ G = Identified the areas with the population density data on QGIS
- ❖ H = Identified the roads and railways on the map
- ❖ I = Identified the water bodies and coastal lines

A. Step-by-step process for the execution of the tasks:

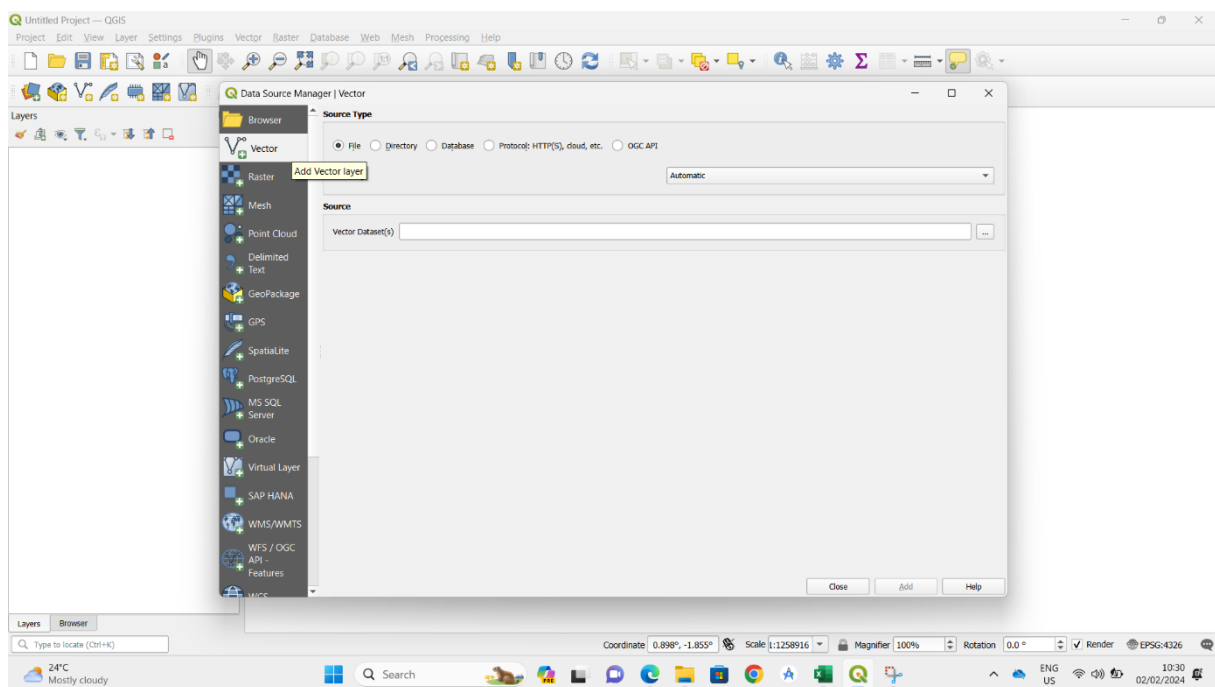
1. Open the QGIS user interface, it should appear as shown on the picture below:



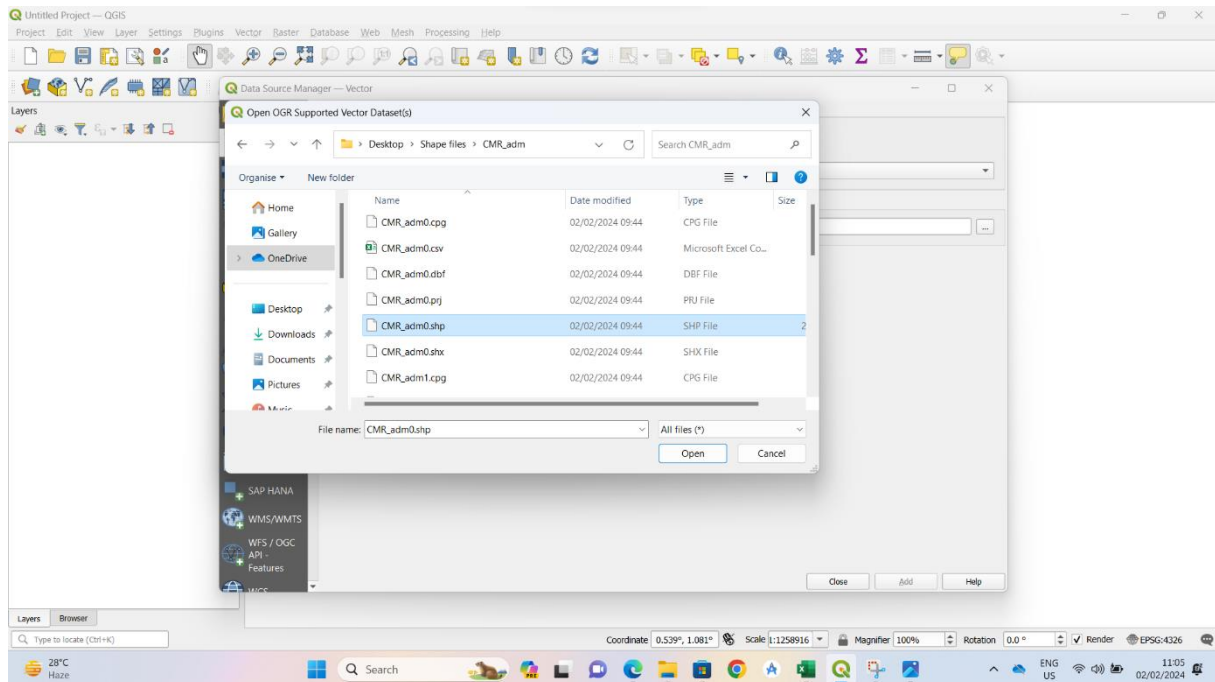
2. Click on the open data source manager to upload the necessary datasets



3. Choose vector on the side bar of the data source manager in order to add a vector layer, the page will appear as shown below:

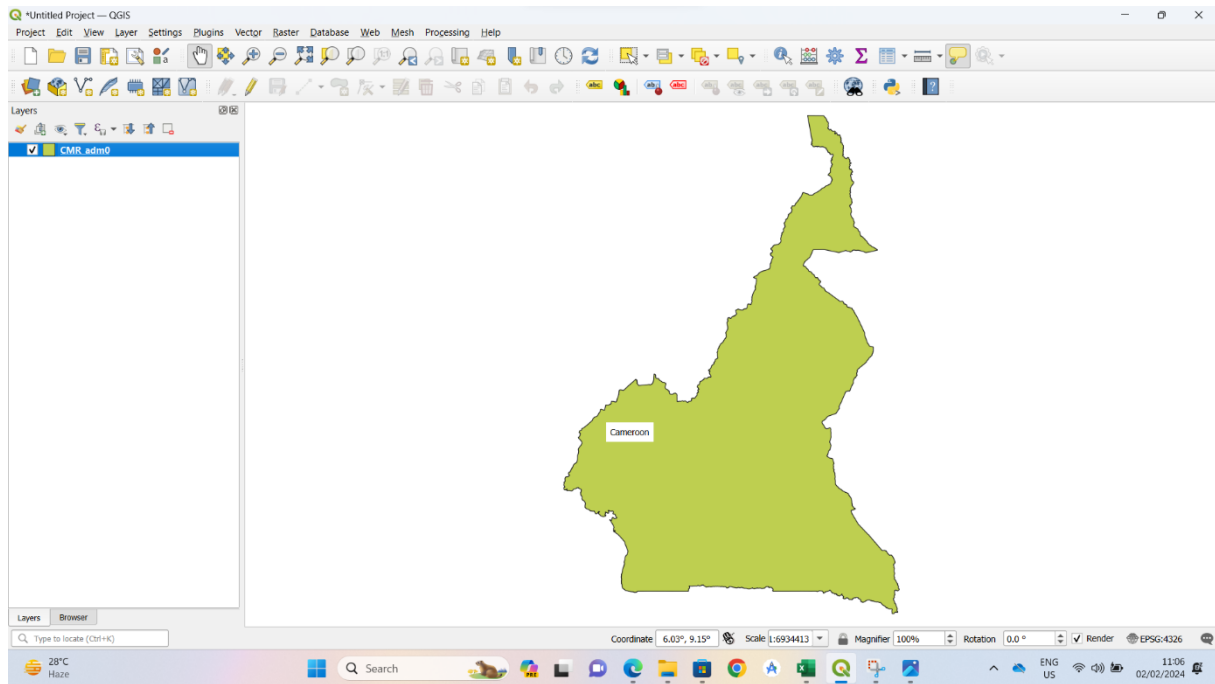


4. Choose source type file and load the vector datasets by clicking on the browse button at the end which is represented with three dots and your different directories should appear as seen on the picture below:

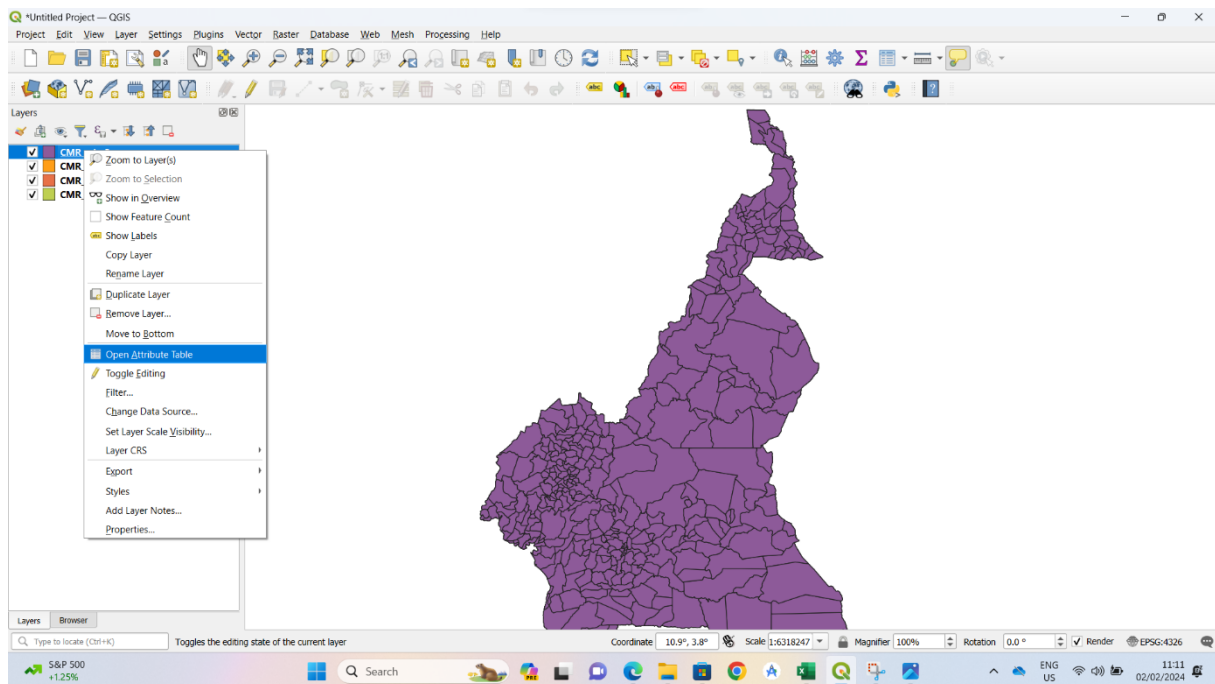


5. Select the file you want to upload (Note that only shapes files with the extension shp can be uploaded into the software) In this case we uploaded the file CMR_adm0.shp

6. After uploading the different files you will see the output displayed on the the map view area. Below is a picture of the result we obtained after uploading our first shp file:



7. We did this for the shp files concerning the **administrative quarters of Cameroon ,the roads, water bodies, coastal lines, railways** we wanted to upload(CMR_adm0,CMR_adm1,CMR_adm2,CMR_adm3,coastallines.shp,cmr_roads.shp,locations which will be seen in the final output) and we simply obtained the same output shown above just that they had different colors.
8. We later on right clicked on the first administrative area which was CMR_adm3 and went to its attribute table as shown below:



9. This attribute table contains the ten regions of Cameroon with their different divisions and sub-divisions as shown on the description below you can scroll down to see all:

QGIS CMR_adm3 — Features Total: 360, Filtered: 360, Selected: 0

	ID_0	ISO	NAME_0	ID_1	NAME_1	ID_2	NAME_2	ID_3	NAME_3	TYPE_3	ENGTYPE_3	NIL_NAME_3	VARNAME_3
1	41	CMR	Cameroon	1	Adamaoua	1	Djerem	1	Ngaoundal	Arondissement	Arondissement	NULL	NULL
2	41	CMR	Cameroon	1	Adamaoua	1	Djerem	2	Tibati	Arondissement	Arondissement	NULL	NULL
3	41	CMR	Cameroon	1	Adamaoua	2	Faro et Déo	3	Galim Tignère	Arondissement	Arondissement	NULL	NULL
4	41	CMR	Cameroon	1	Adamaoua	2	Faro et Déo	4	Mayo Baléo	Arondissement	Arondissement	NULL	NULL
5	41	CMR	Cameroon	1	Adamaoua	2	Faro et Déo	5	Tignère	Arondissement	Arondissement	NULL	NULL
6	41	CMR	Cameroon	1	Adamaoua	3	Mayo Banyo	6	Bankim	Arondissement	Arondissement	NULL	NULL
7	41	CMR	Cameroon	1	Adamaoua	3	Mayo Banyo	7	Banyo	Arondissement	Arondissement	NULL	NULL
8	41	CMR	Cameroon	1	Adamaoua	3	Mayo Banyo	8	Mayo Darlé	Arondissement	Arondissement	NULL	NULL
9	41	CMR	Cameroon	1	Adamaoua	4	Mbéré	9	Dir	Arondissement	Arondissement	NULL	NULL
10	41	CMR	Cameroon	1	Adamaoua	4	Mbéré	10	Djohong	Arondissement	Arondissement	NULL	NULL
11	41	CMR	Cameroon	1	Adamaoua	4	Mbéré	11	Meiganga	Arondissement	Arondissement	NULL	NULL
12	41	CMR	Cameroon	1	Adamaoua	5	Vina	12	Belel	Arondissement	Arondissement	NULL	NULL
13	41	CMR	Cameroon	1	Adamaoua	5	Vina	13	Martap	Arondissement	Arondissement	NULL	NULL
14	41	CMR	Cameroon	1	Adamaoua	5	Vina	14	Mbé	Arondissement	Arondissement	NULL	NULL
15	41	CMR	Cameroon	1	Adamaoua	5	Vina	15	Nganha	Arondissement	Arondissement	NULL	NULL
16	41	CMR	Cameroon	1	Adamaoua	5	Vina	16	Ngaoundéré 1	Arondissement	Arondissement	NULL	NULL
17	41	CMR	Cameroon	1	Adamaoua	5	Vina	17	Ngaoundéré 2	Arondissement	Arondissement	NULL	NULL
18	41	CMR	Cameroon	1	Adamaoua	5	Vina	18	Ngaoundéré 3	Arondissement	Arondissement	NULL	NULL
19	41	CMR	Cameroon	1	Adamaoua	5	Vina	19	Nyambaka	Arondissement	Arondissement	NULL	NULL
20	41	CMR	Cameroon	2	Centre	6	Haute Sanaga	20	Bibé	Arondissement	Arondissement	NULL	NULL
21	41	CMR	Cameroon	2	Centre	6	Haute Sanaga	21	Lembé Yéoum	Arondissement	Arondissement	NULL	NULL
22	41	CMR	Cameroon	2	Centre	6	Haute Sanaga	22	Mbandjock	Arondissement	Arondissement	NULL	NULL
23	41	CMR	Cameroon	2	Centre	6	Haute Sanaga	23	Minta	Arondissement	Arondissement	NULL	NULL
24	41	CMR	Cameroon	2	Centre	6	Haute Sanaga	24	Nanga Eboko	Arondissement	Arondissement	NULL	NULL

Show All Features

28°C Haze

Search

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10. We known proceeded to the selection of the different regions on the map in other to differentiate them by implementing unique colors on each of them. We started by selecting the Adamawa region as shown below:

QGIS CMR_adm3 — Features Total: 360, Filtered: 360, Selected: 19

	ID_0	ISO	NAME_0	ID_1	NAME_1	ID_2	NAME_2	ID_3	NAME_3	TYPE_3	ENGTYPE_3	NIL_NAME_3	VARNAME_3
1	41	CMR	Cameroon	1	Adamaoua	1	Djerem	1	Ngaoundal	Arondissement	Arondissement	NULL	NULL
2	41	CMR	Cameroon	1	Adamaoua	1	Djerem	2	Tibati	Arondissement	Arondissement	NULL	NULL
3	41	CMR	Cameroon	1	Adamaoua	2	Faro et Déo	3	Galim Tignère	Arondissement	Arondissement	NULL	NULL
4	41	CMR	Cameroon	1	Adamaoua	2	Faro et Déo	4	Mayo Baléo	Arondissement	Arondissement	NULL	NULL
5	41	CMR	Cameroon	1	Adamaoua	2	Faro et Déo	5	Tignère	Arondissement	Arondissement	NULL	NULL
6	41	CMR	Cameroon	1	Adamaoua	3	Mayo Banyo	6	Bankim	Arondissement	Arondissement	NULL	NULL
7	41	CMR	Cameroon	1	Adamaoua	3	Mayo Banyo	7	Banyo	Arondissement	Arondissement	NULL	NULL
8	41	CMR	Cameroon	1	Adamaoua	3	Mayo Banyo	8	Mayo Darlé	Arondissement	Arondissement	NULL	NULL
9	41	CMR	Cameroon	1	Adamaoua	4	Mbéré	9	Dir	Arondissement	Arondissement	NULL	NULL
10	41	CMR	Cameroon	1	Adamaoua	4	Mbéré	10	Djohong	Arondissement	Arondissement	NULL	NULL
11	41	CMR	Cameroon	1	Adamaoua	4	Mbéré	11	Meiganga	Arondissement	Arondissement	NULL	NULL
12	41	CMR	Cameroon	1	Adamaoua	5	Vina	12	Belel	Arondissement	Arondissement	NULL	NULL
13	41	CMR	Cameroon	1	Adamaoua	5	Vina	13	Martap	Arondissement	Arondissement	NULL	NULL
14	41	CMR	Cameroon	1	Adamaoua	5	Vina	14	Mbé	Arondissement	Arondissement	NULL	NULL
15	41	CMR	Cameroon	1	Adamaoua	5	Vina	15	Nganha	Arondissement	Arondissement	NULL	NULL
16	41	CMR	Cameroon	1	Adamaoua	5	Vina	16	Ngaoundéré 1	Arondissement	Arondissement	NULL	NULL
17	41	CMR	Cameroon	1	Adamaoua	5	Vina	17	Ngaoundéré 2	Arondissement	Arondissement	NULL	NULL
18	41	CMR	Cameroon	1	Adamaoua	5	Vina	18	Ngaoundéré 3	Arondissement	Arondissement	NULL	NULL
19	41	CMR	Cameroon	1	Adamaoua	5	Vina	19	Nyambaka	Arondissement	Arondissement	NULL	NULL
20	41	CMR	Cameroon	2	Centre	6	Haute Sanaga	20	Bibé	Arondissement	Arondissement	NULL	NULL
21	41	CMR	Cameroon	2	Centre	6	Haute Sanaga	21	Lembé Yéoum	Arondissement	Arondissement	NULL	NULL
22	41	CMR	Cameroon	2	Centre	6	Haute Sanaga	22	Mbandjock	Arondissement	Arondissement	NULL	NULL
23	41	CMR	Cameroon	2	Centre	6	Haute Sanaga	23	Minta	Arondissement	Arondissement	NULL	NULL
24	41	CMR	Cameroon	2	Centre	6	Haute Sanaga	24	Nanga Eboko	Arondissement	Arondissement	NULL	NULL

Show All Features

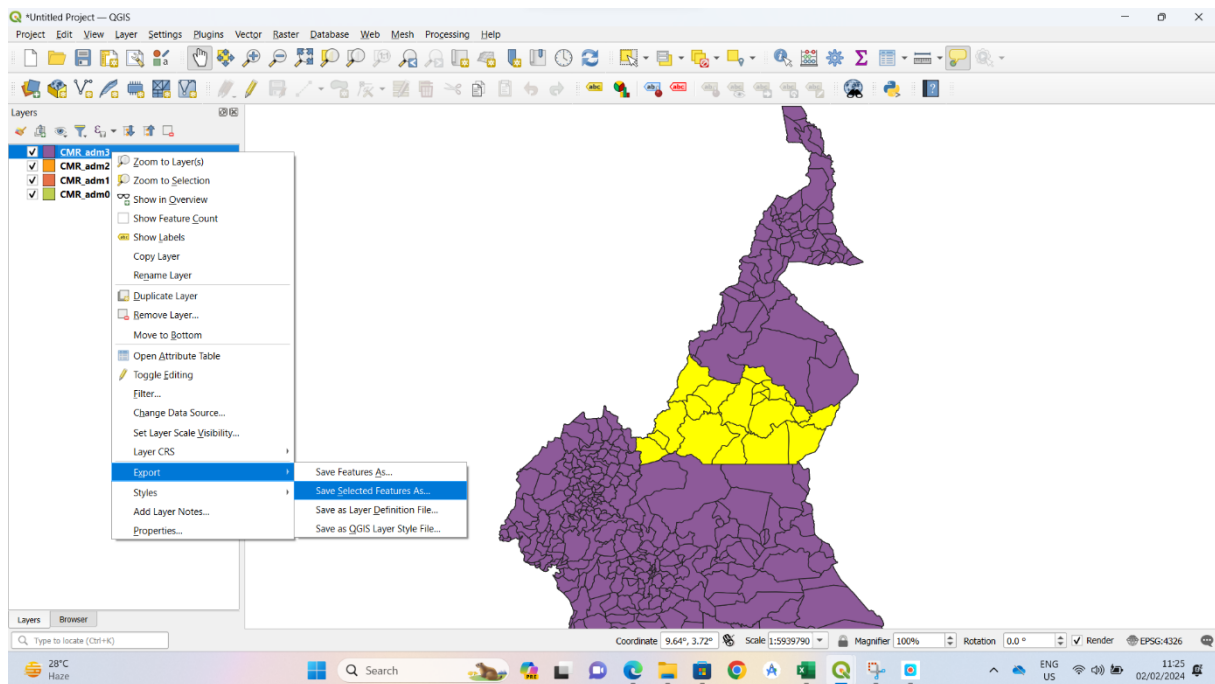
28°C Haze

Search

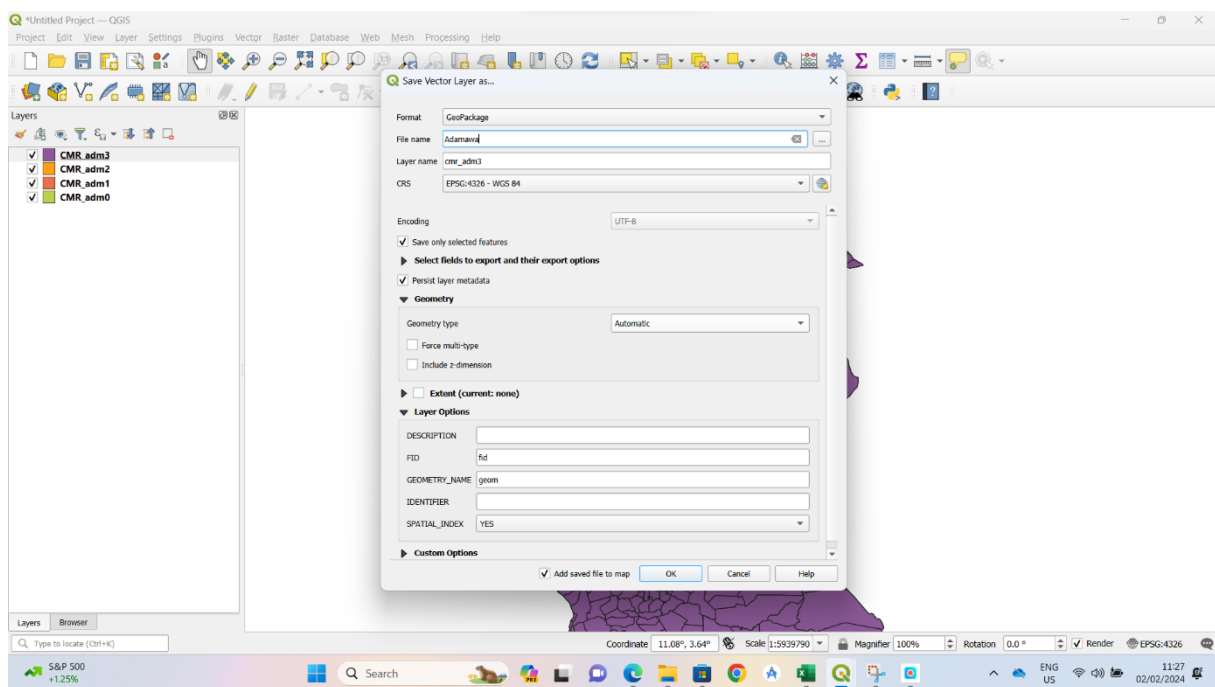
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As you can see from the picture above we selected just the features that belonged to the Adamawa region

11. We then close this page and we will notice that the portion of the selected area has already changed colour with a default color, right clicked back on CMR_adm3 then we **choosed export>then save selected Feature as>** this will be seen on the picture below:

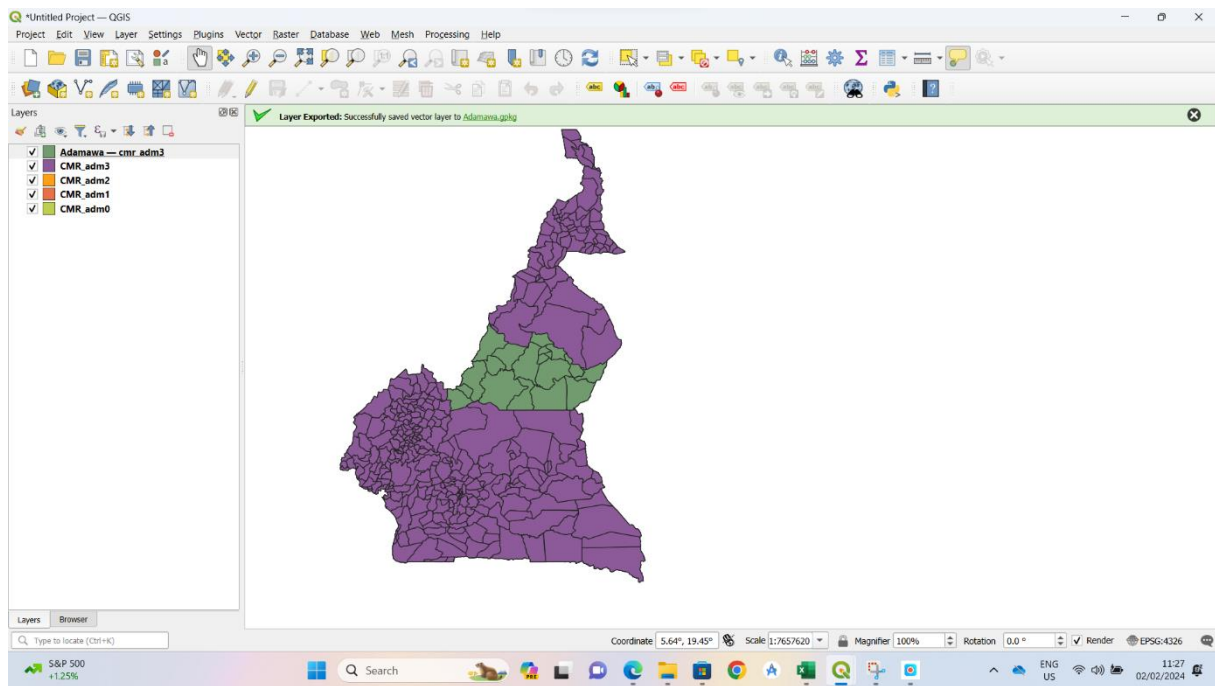


12. As soon as we click on it we will see the above image being displayed:

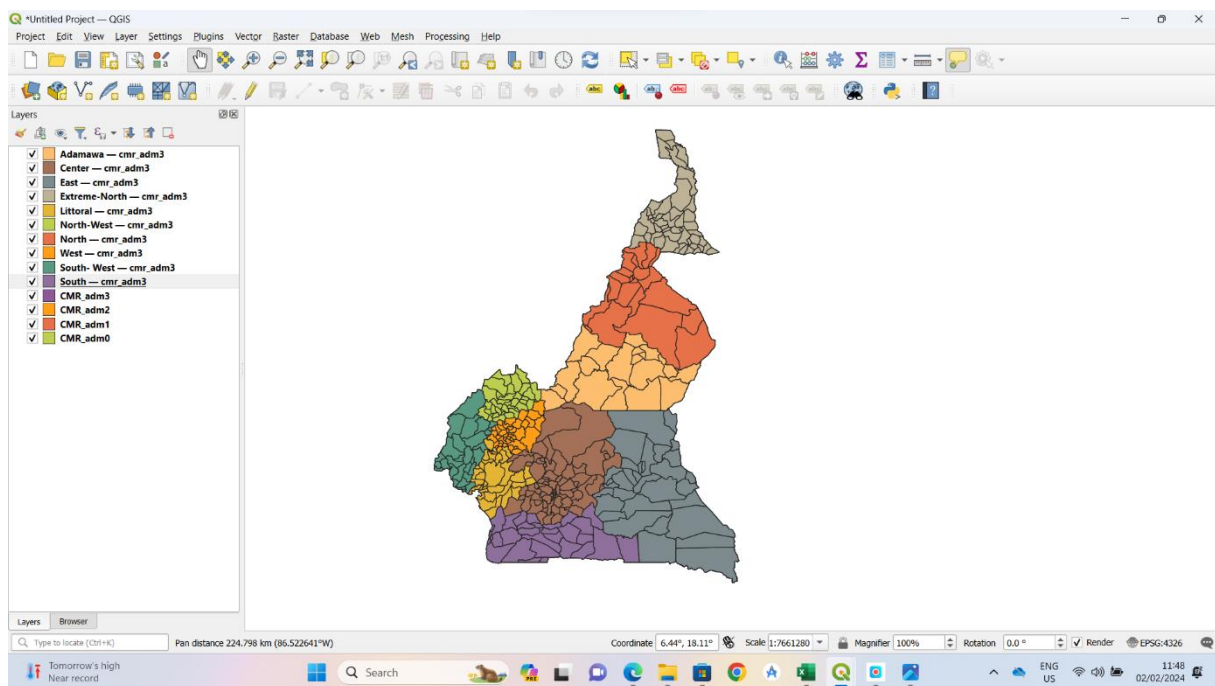


Here we will simply enter a name where we have File name in order to give a name to the selected feature we want to save . In our case we named the file as Adamawa.

13. Then clicked on OK to apply the changes made . After this if you notice well you will see a new layer been added to the layers panel. **Note that** you can decide to change the color of the selected area by right clicking on the layer in the layers pannel and choosing properties there you can known decide to change the colour. For our case we known had the image below:

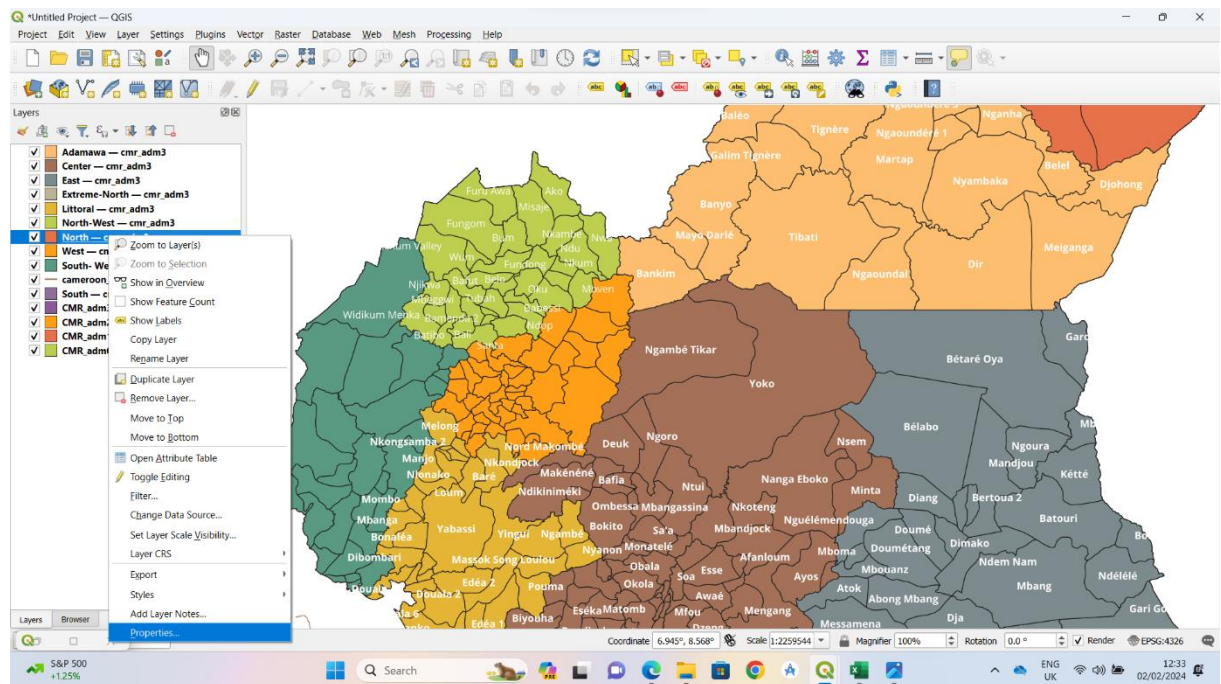


14. Known repeat this step for any particular zone on the map you want to identify easily. In our case we did this for the ten regions of Cameroon that is we repeated the step above 10 times and we had the output below:

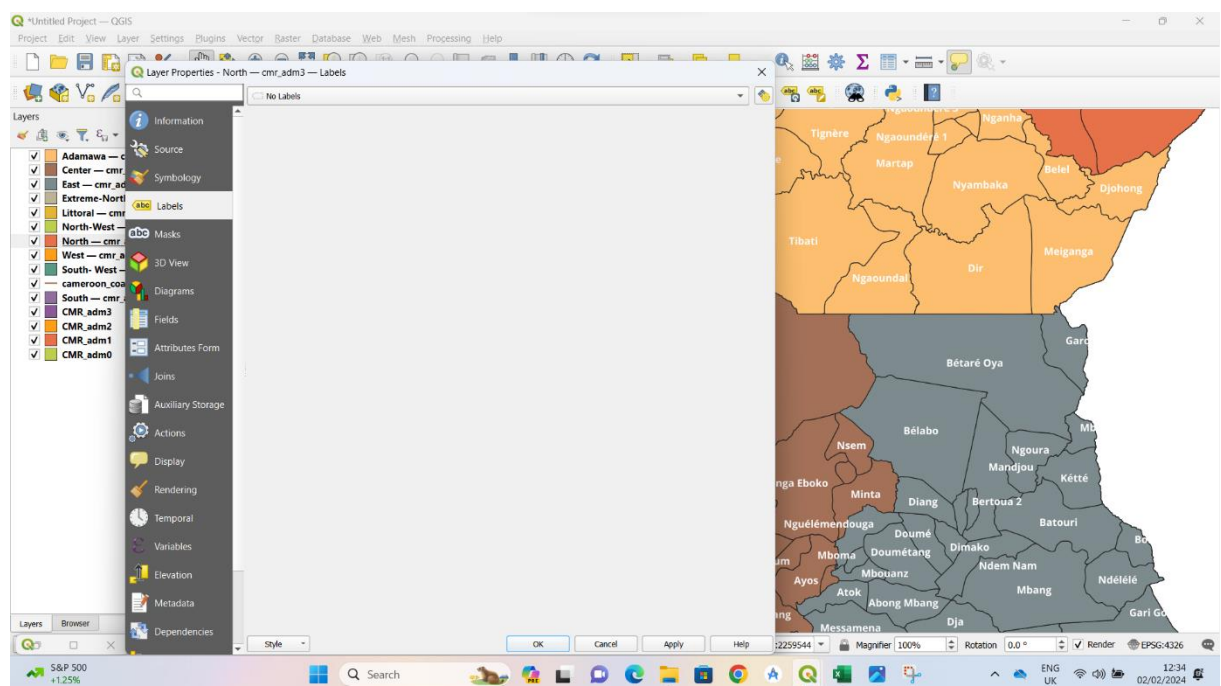


If you notice well on the layers pannel we have the labels of the 10 regions of Cameroon each having a unique colour.

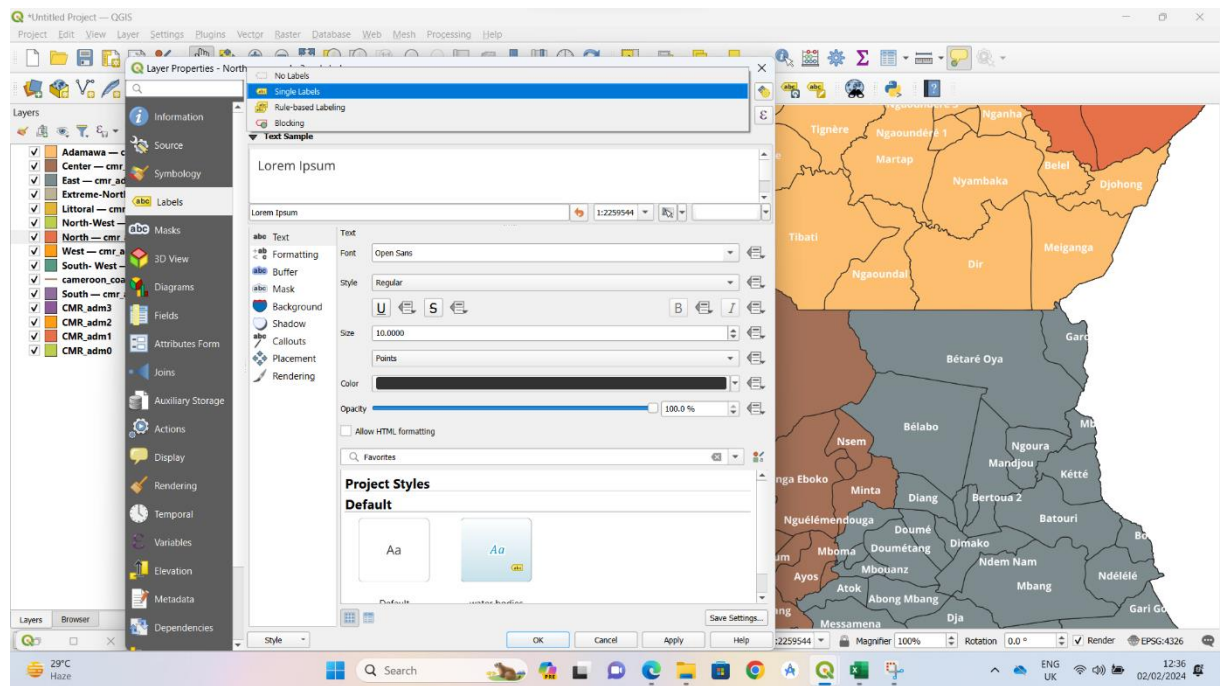
15. As you can see on the map above we don't have the different department , towns .etc of this region so to identify them and bring them on the map we will right click on the layer of each region on the layer panel and click on properties as shown on the picture below:



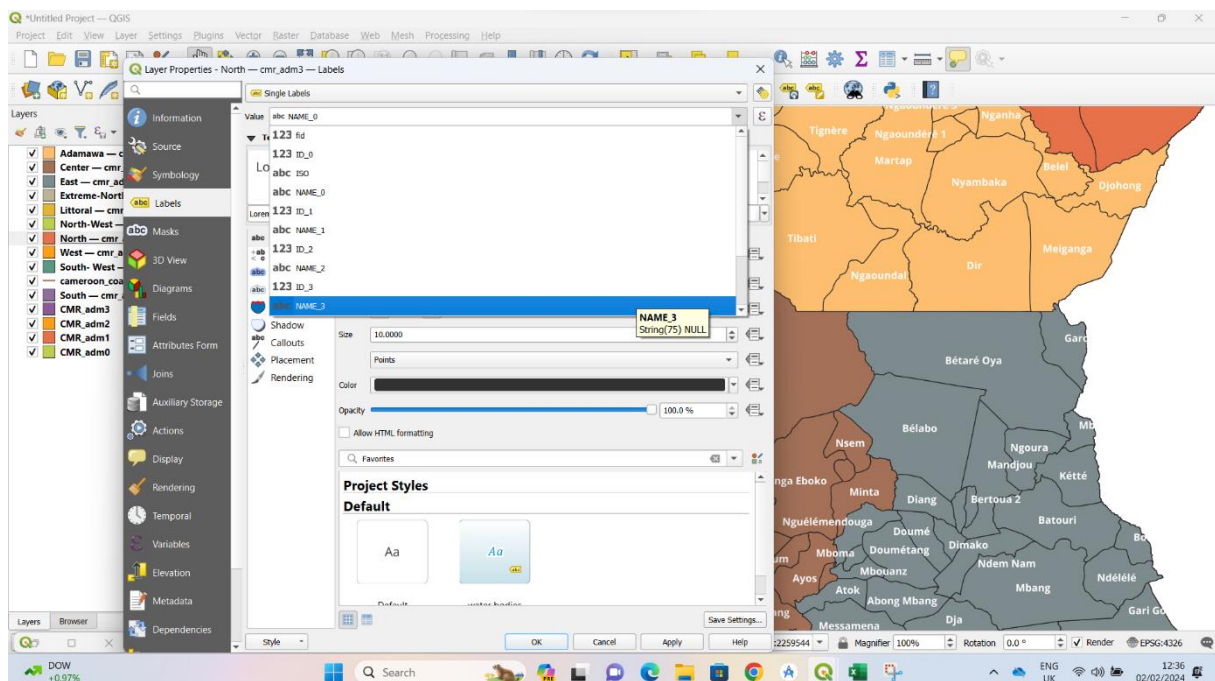
16. After clicking on properties as shown on the picture above you will have a page displayed to you where you will see a side bar that contains different options, you then click on the option labels(it has a logo with abc) as shown on the picture below:



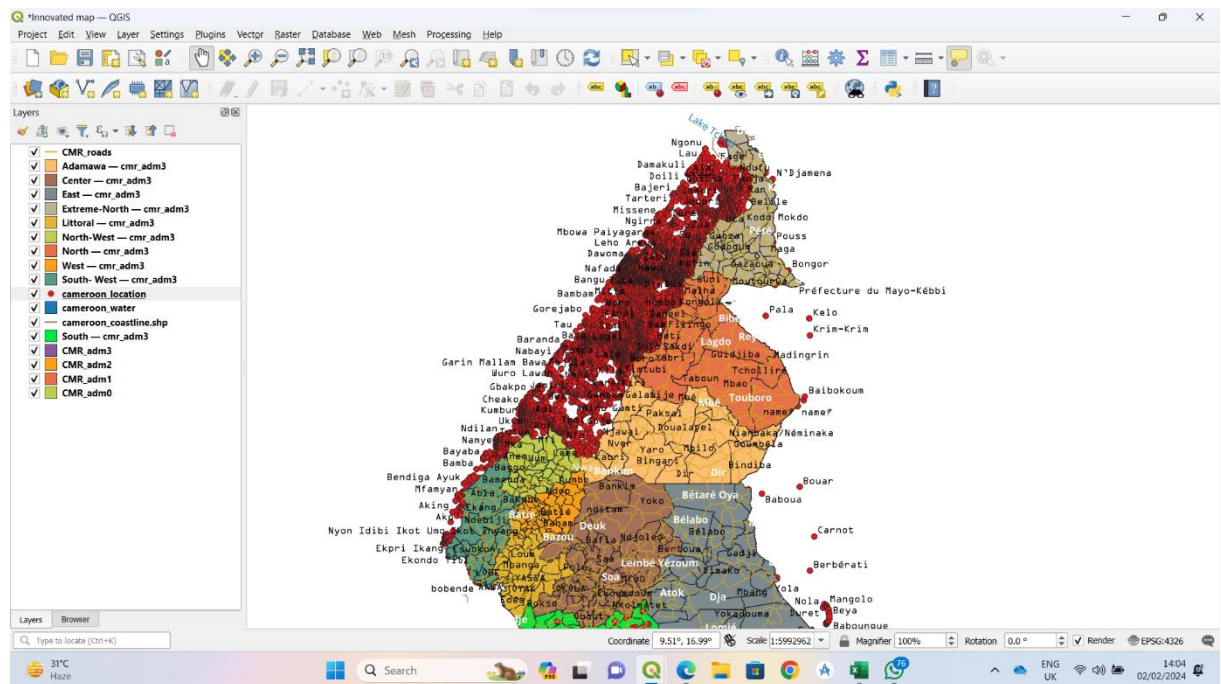
17. After clicking on the option Labels a page will be displayed as shown on the image above. The page contains a label named “No Labels” which has a dropdown button at the end of the label, click on that dropdown button and choose single labels then a form will be displayed as shown below:



18. On the label form displayed above you will have to click on value and choose the title that contains the data you want it to appear on the map . As for this our case we chose NAME_3 (for the North Region) because we want the data it contained to be displayed on the map. Later on with the other attributes this form contains you can perform modifications such choosing how you will want the labels to appear(bold,italic etc) then you can click on apply then OK:



19. After carrying out the steps above the labels will be displayed as shown below:



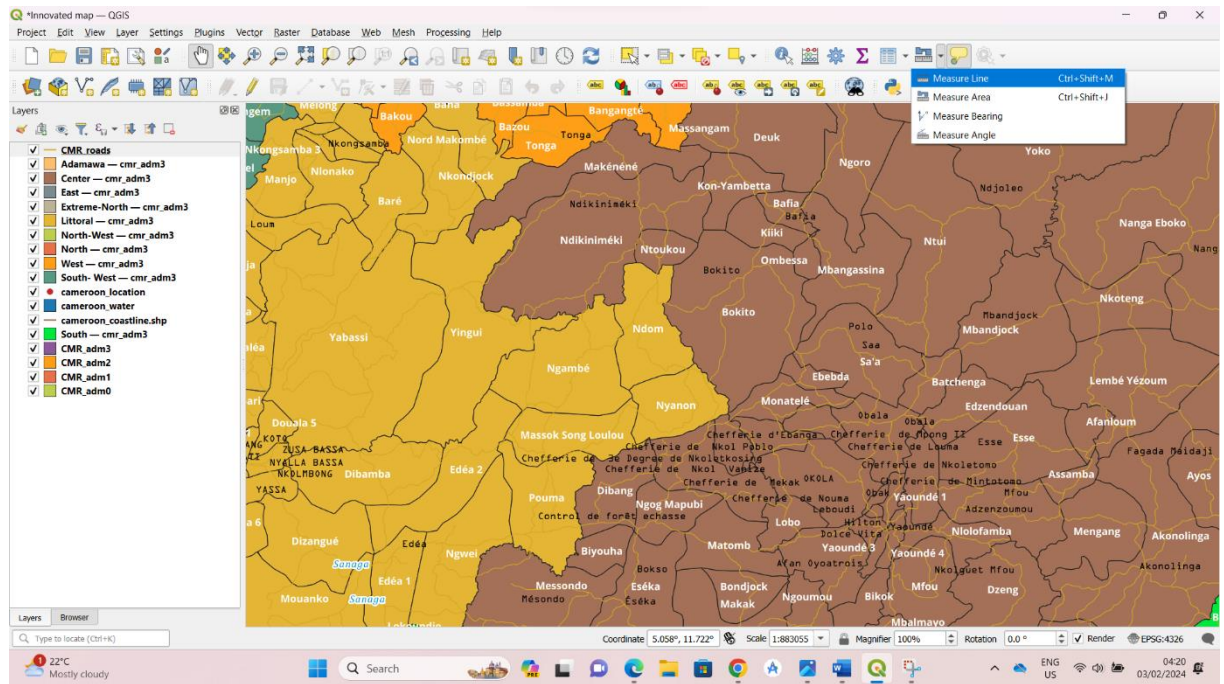
20. Please note that I applied the labels just for the North Region, so what you see on this image above is just an output of what I obtained after applying the labels of the 10 Regions of Cameroon

21. On the above image we can see the different shp files that were uploaded unto the layer panel we include:

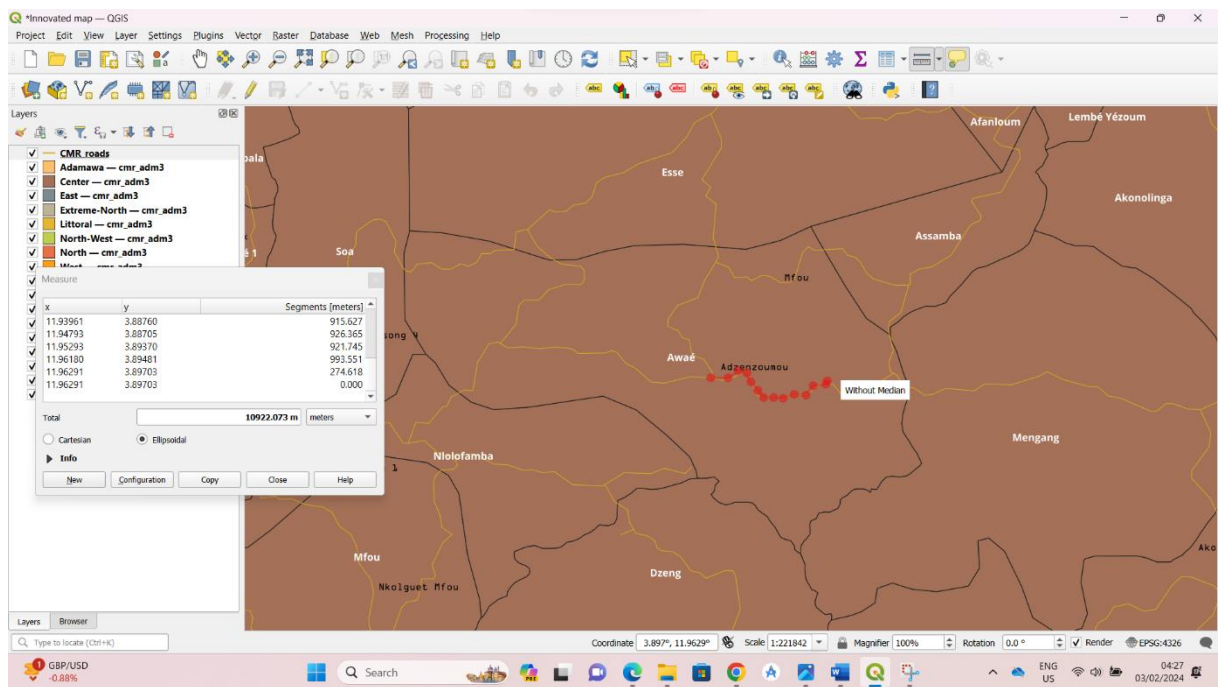
- The 10 regions of Cameroon its divisions and sub-divisions
- Cameroon water bodies
- Different location points in Cameroon
- Coastal lines
- Roads
- Railways

Known on the available map view above we will give a step by step demonstration of how calculations can be made on the map precisely measuring a line:

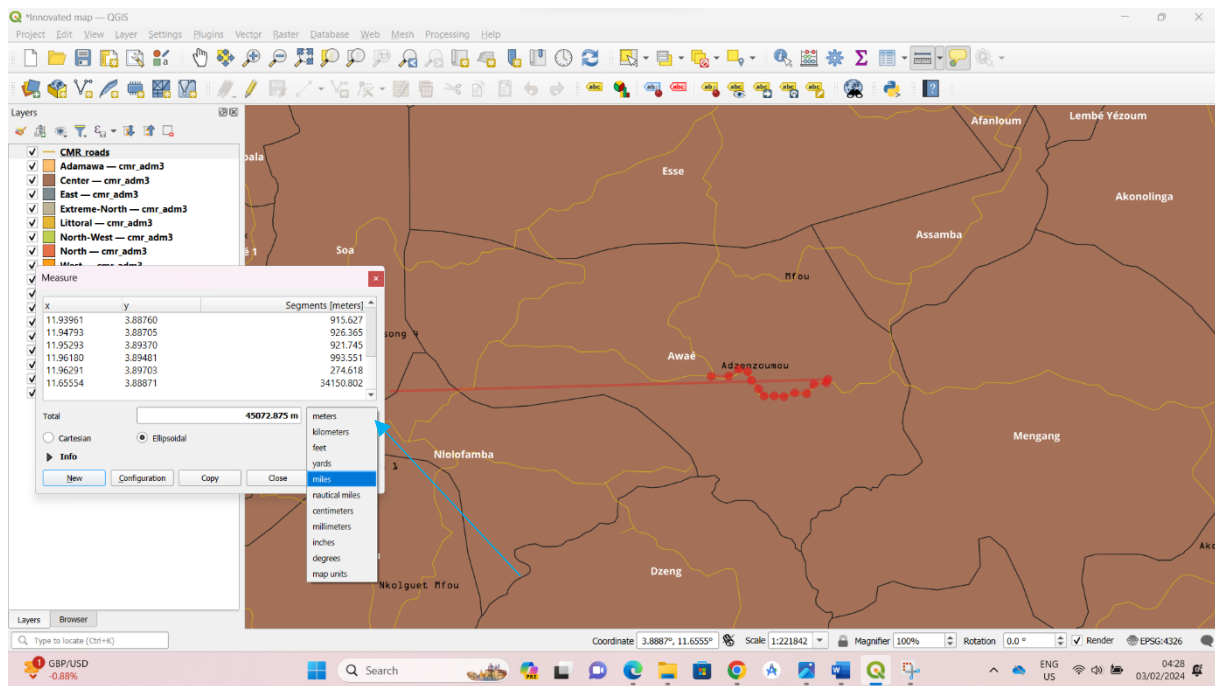
- Go to your attributes tool bar and click on the icon thaht has a ruler where you see Measure Area in other to carry out your measurements on the map as shown below:



On the above image we chose measure line in order to measure the distance between two points as shown on this image below:

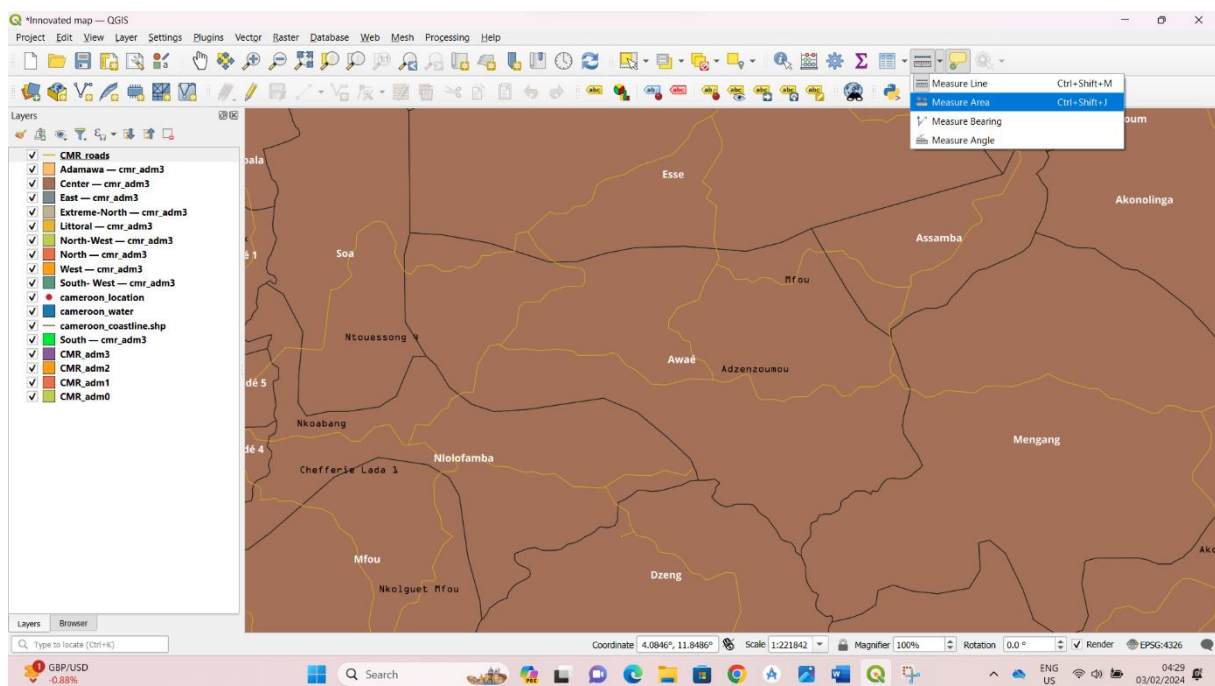


As seen on this image above where you see those red dots is the line I selected in order to calculate its length. After calculation this white box in the image shows the length of the selected area in the label total. And note that this total is not just calculated in meters as shown by default but can be calculated also in many different units of measurement as shown on the picture below:

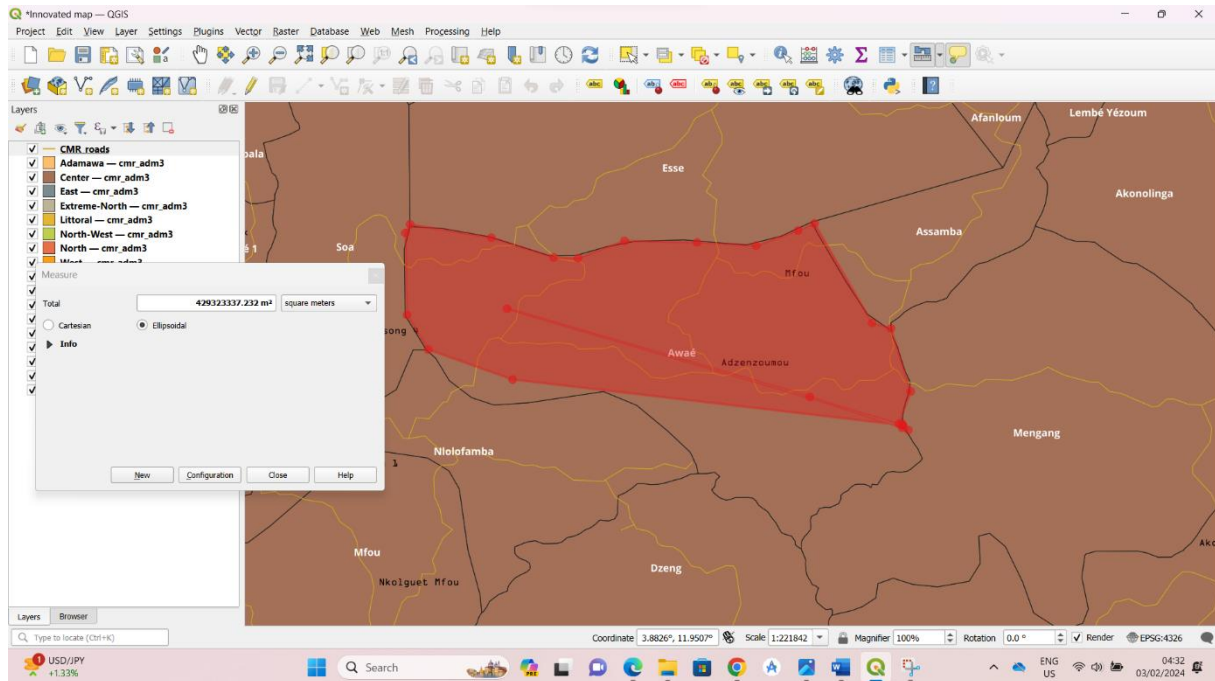


Known you can measure the surface area on the map by following the following step:

- Click on the measure area tool found on the attribute tool bar and choose measure area as shown on the image below:



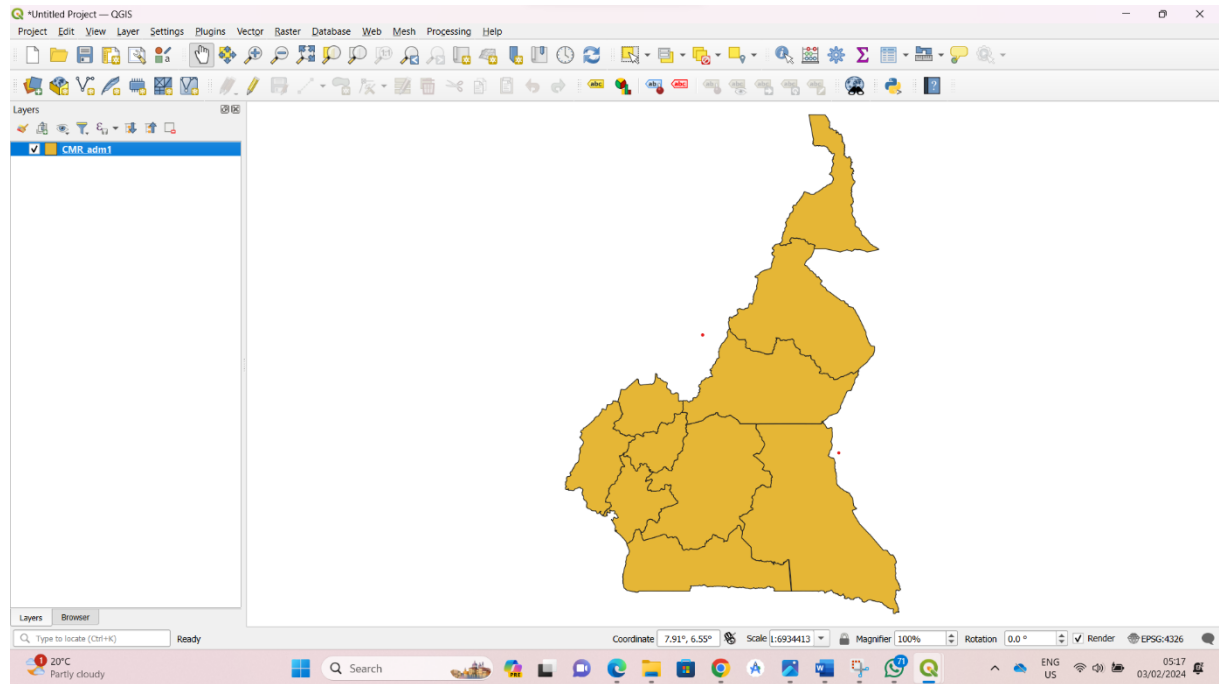
- After clicking on it a white box will be displayed and you can know select the area you want to calculate its surface area as shown on the image below:



The selected area in red is the portion I chose to calculate it's area and it's area is seen in the white box aside and its unit of measurement can be changed.

Known we will talk about how we can represent the population Density in Cameroon step by step:

1. Lets just say you have already opened QGIS load in your Cameroon map (the administrative map that contains the ten regions of Cameroon in our case is CMR_adm1)then it shall appear as shown below:

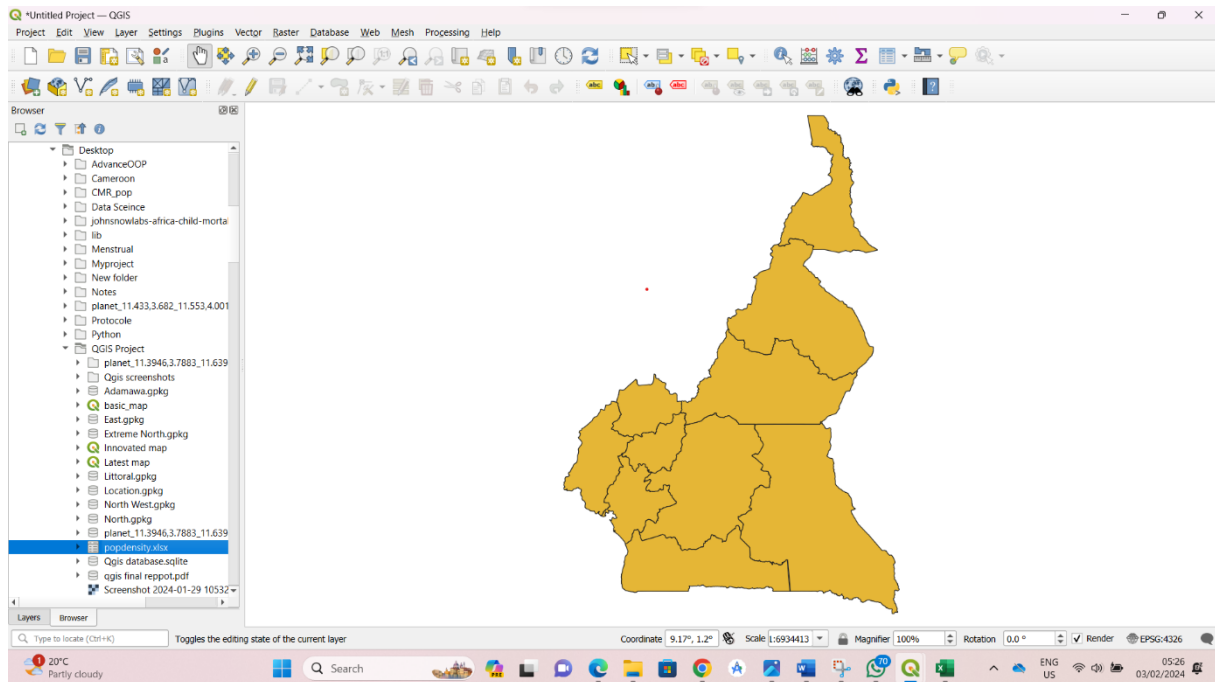


2. Make sure your Excell file that contains surface area and the population of the 10 regions of cameroon is already ready. For our case this is it below:

The screenshot shows an Excel spreadsheet with the following data:

Name	Area	Population
Adamaoua	63,701	1,201,000
Centre	68,953	4,159,500
Est	109,002	835,600
Extrême-Nord	34,263	3,993,000
Littoral	20,248	3,355,000
Nord-Ouest	17,300	1,968,600
Nord	66,090	2,442,600
Ouest	13,892	1,921,600
Sud-Ouest	25,410	1,553,300
Sud	47,191	749,600

- Open your browser panel and add your excel file to the layer panel for our case the excel file was called popdensity.xlsx as shown below:

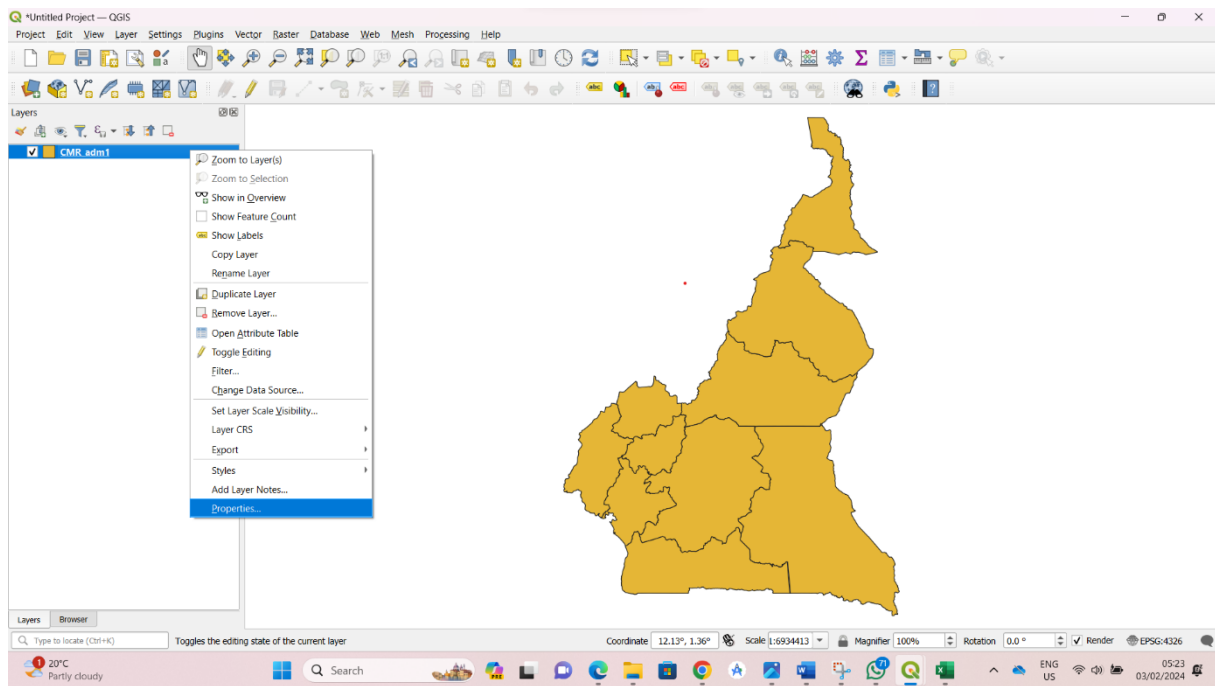


- Open your attribute table for your CMR_adm1 layer by right clicking on it and choosing open attribute tables for it to display you the table below for you to make sure the data it contains is that of the ten regions of cameroon:

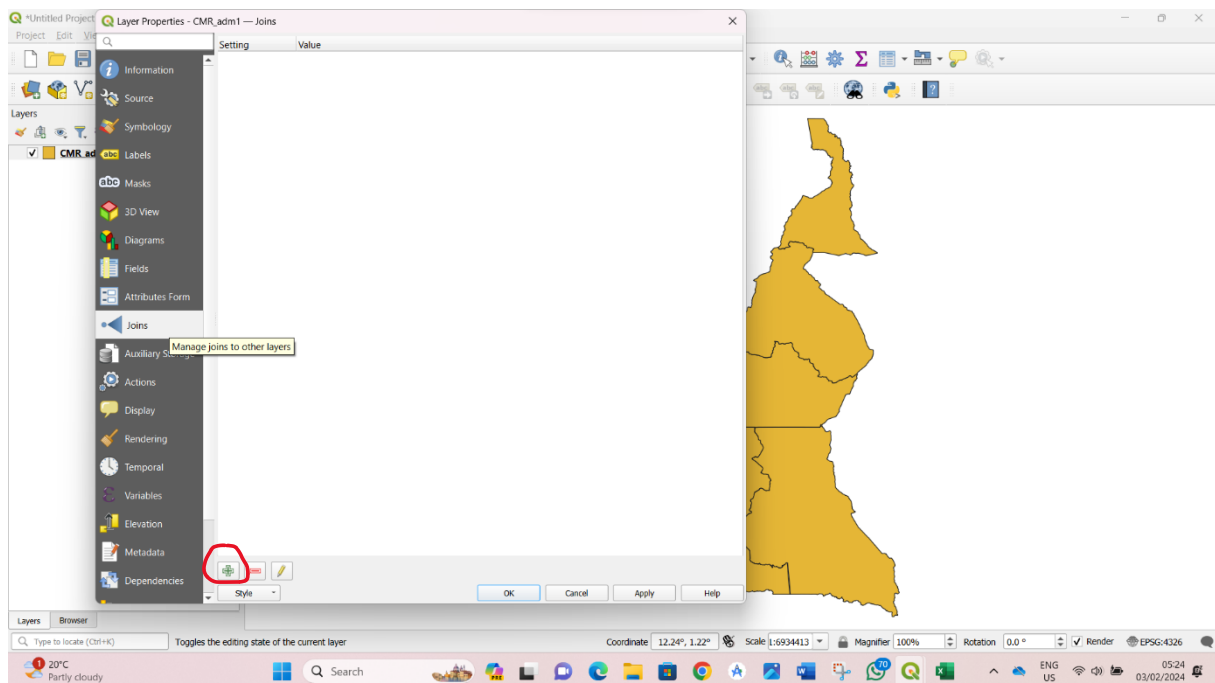
CMR_adm1 — Features Total: 10, Filtered: 10, Selected: 0

	ID_0	ISO	NAME_0	ID_1	NAME_1	TYPE_1	ENGTYPE_1	NL_NAME_1	VARNAME_1
1	41	CMR	Cameroon	1	Adamaoua	Région	Region	NULL	Adamoua(Ada...
2	41	CMR	Cameroon	2	Centre	Région	Region	NULL	Central
3	41	CMR	Cameroon	3	Est	Région	Region	NULL	East
4	41	CMR	Cameroon	4	Extrême-Nord	Région	Region	NULL	Extreme-North
5	41	CMR	Cameroon	5	Littoral	Région	Region	NULL	Litoral
6	41	CMR	Cameroon	6	Nord-Ouest	Région	Region	NULL	North-West
7	41	CMR	Cameroon	7	Nord	Région	Region	NULL	North
8	41	CMR	Cameroon	8	Ouest	Région	Region	NULL	West
9	41	CMR	Cameroon	9	Sud-Ouest	Région	Region	NULL	South-West
10	41	CMR	Cameroon	10	Sud	Région	Region	NULL	South

- Known right click on your layer and choose properties to open the properties of this layer as shown below:

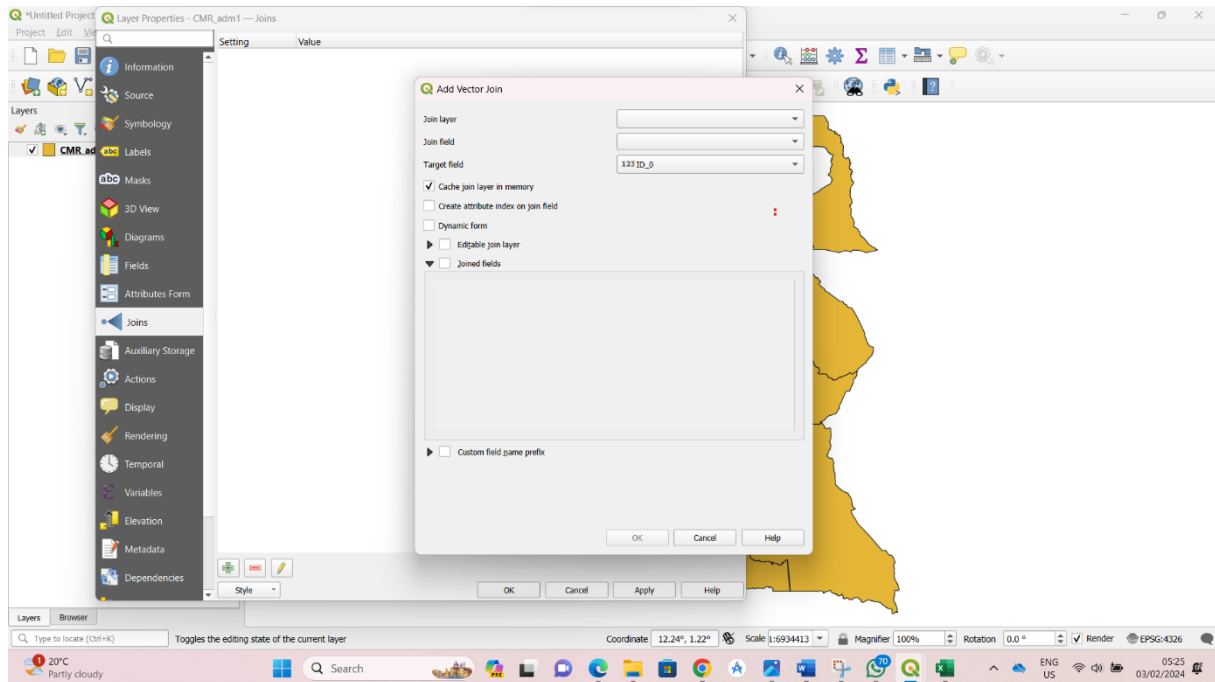


6. After the page below has been displayed look for joins on the side bar and click on it then know click on the green plus sign that is circled in red as shown below:

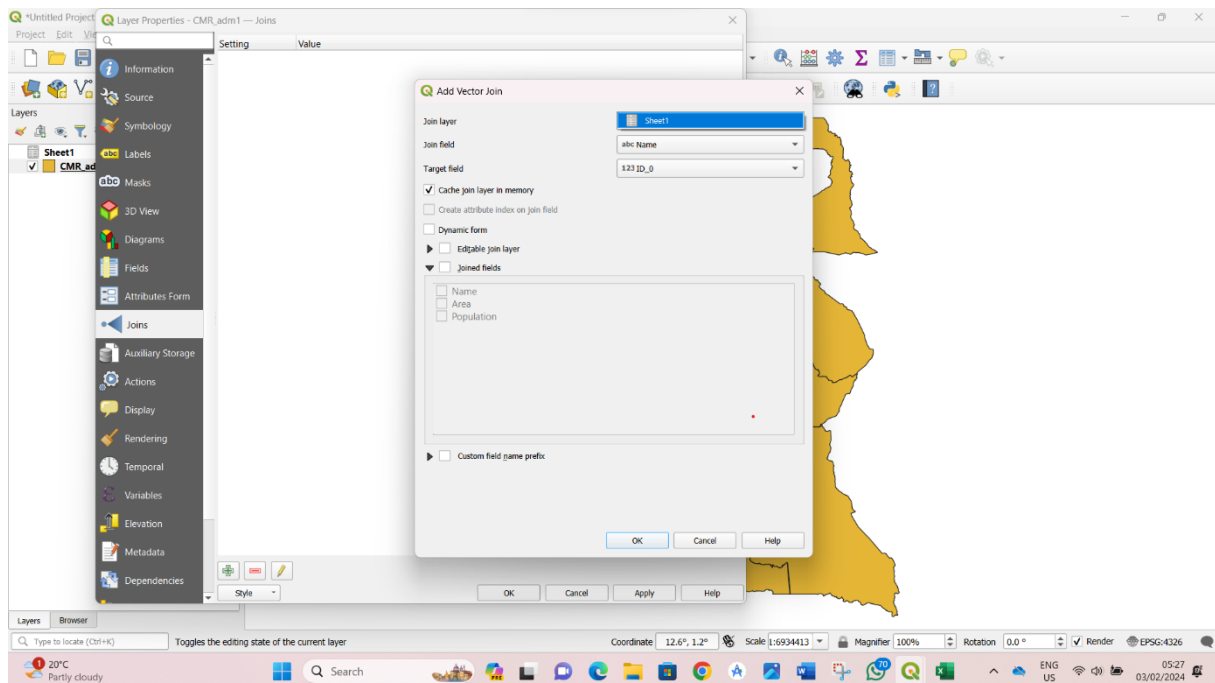


7. Then a form will be displayed where you have to select your excel file you previously uploaded in the Join Layer label and you will have to choose the column that will help you

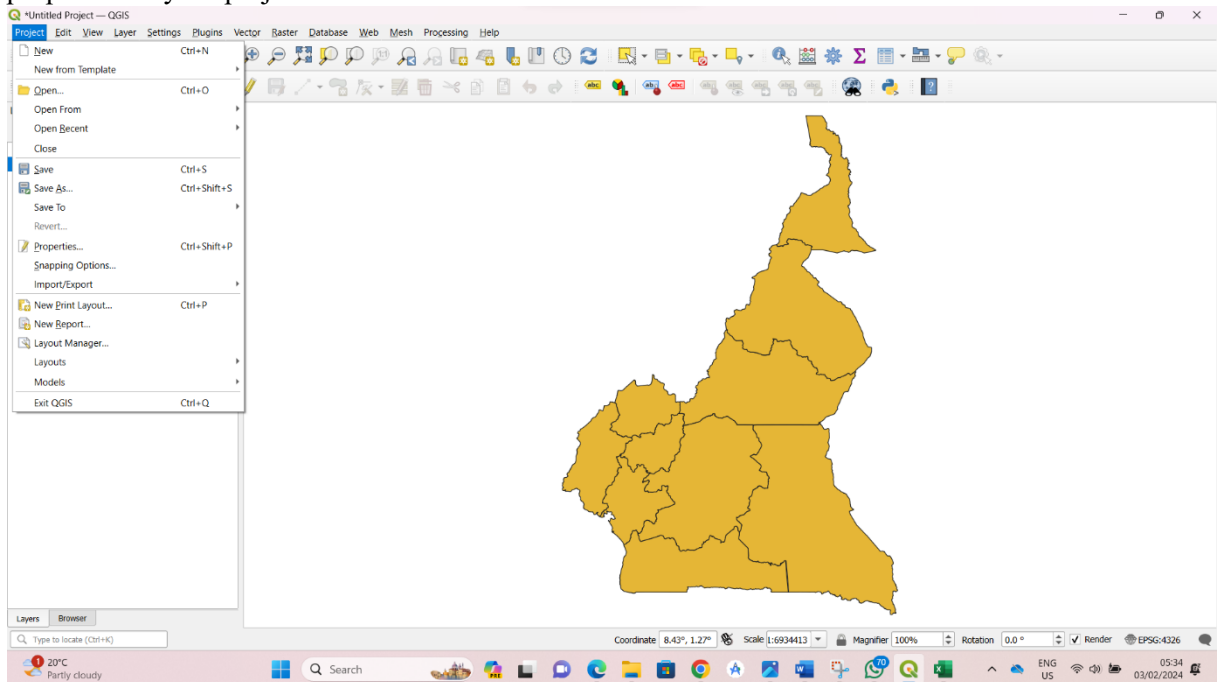
join making sure that the data in that column is the same with the data in your excel file under that column



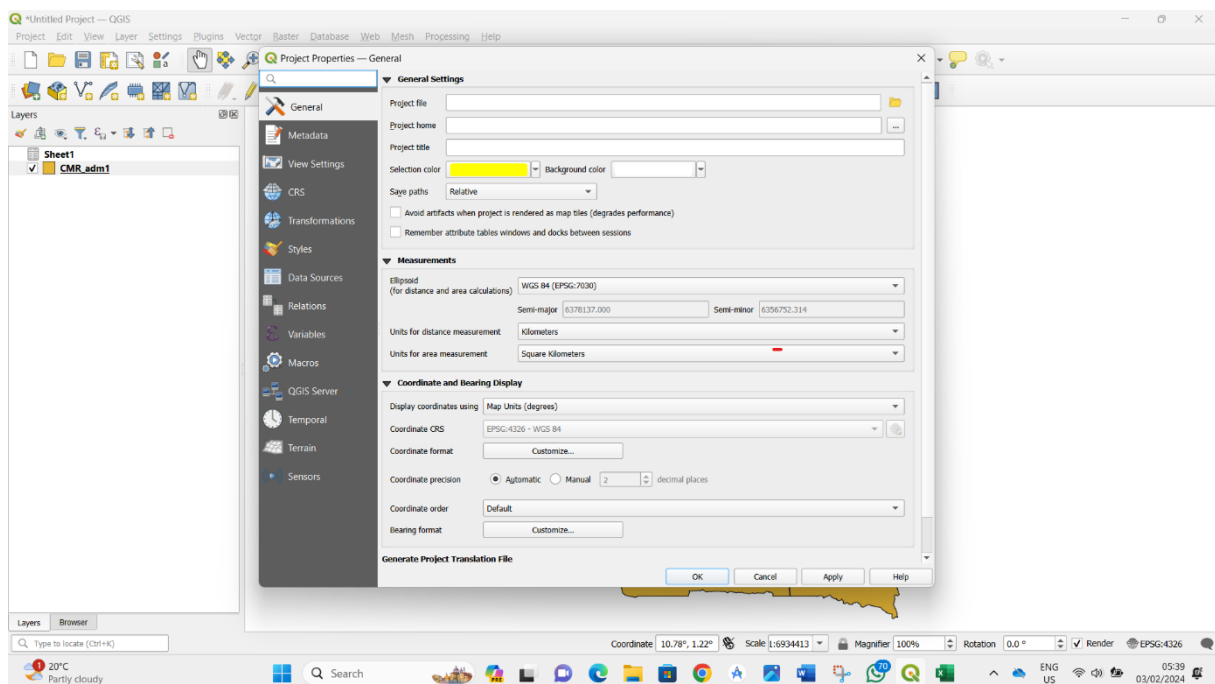
8. As seen on the diagram above the Join layer which is our excel file was added and the joinfield was chosen then after that just press apply the okay.



9. Known go to your menu bar and choose project then known click on properties to check the properties of your project:



10. When you have this page change your units of measurement to kilometers and square kilometers and click on apply then OK:



11. Known as shown below open your attribute table and click on the toggle editing mode:

QGIS - CMR_admin1 - Features Total: 10, Filtered: 10, Selected: 0

Toggle editing mode (Ctrl+E)

Update AllUpdate Selected

ID_0	ISO	NAME_0	ID_1	NAME_1	TYPE_1	ENGTYPE_1	NI_NAME_1	VARNAME_1	Sheet1_Area	Sheet1_Population	
1	41	CMR	Cameroon	1	Adamaoua	Région	Region	NULL	AdamouajAda...	63701	1201000
2	41	CMR	Cameroon	2	Centre	Région	Region	NULL	Central	68953	4159500
3	41	CMR	Cameroon	3	Est	Région	Region	NULL	East	109002	835600
4	41	CMR	Cameroon	4	Extrême-Nord	Région	Region	NULL	Extreme-North	34263	3993000
5	41	CMR	Cameroon	5	Littoral	Région	Region	NULL	Litoral	20248	3355000
6	41	CMR	Cameroon	6	Nord-Ouest	Région	Region	NULL	North-West	17300	1968600
7	41	CMR	Cameroon	7	Nord	Région	Region	NULL	North	66090	2442600
8	41	CMR	Cameroon	8	Ouest	Région	Region	NULL	West	13892	1921600
9	41	CMR	Cameroon	9	Sud-Ouest	Région	Region	NULL	South-West	25410	1553300
10	41	CMR	Cameroon	10	Sud	Région	Region	NULL	South	47191	749600

Show All Features

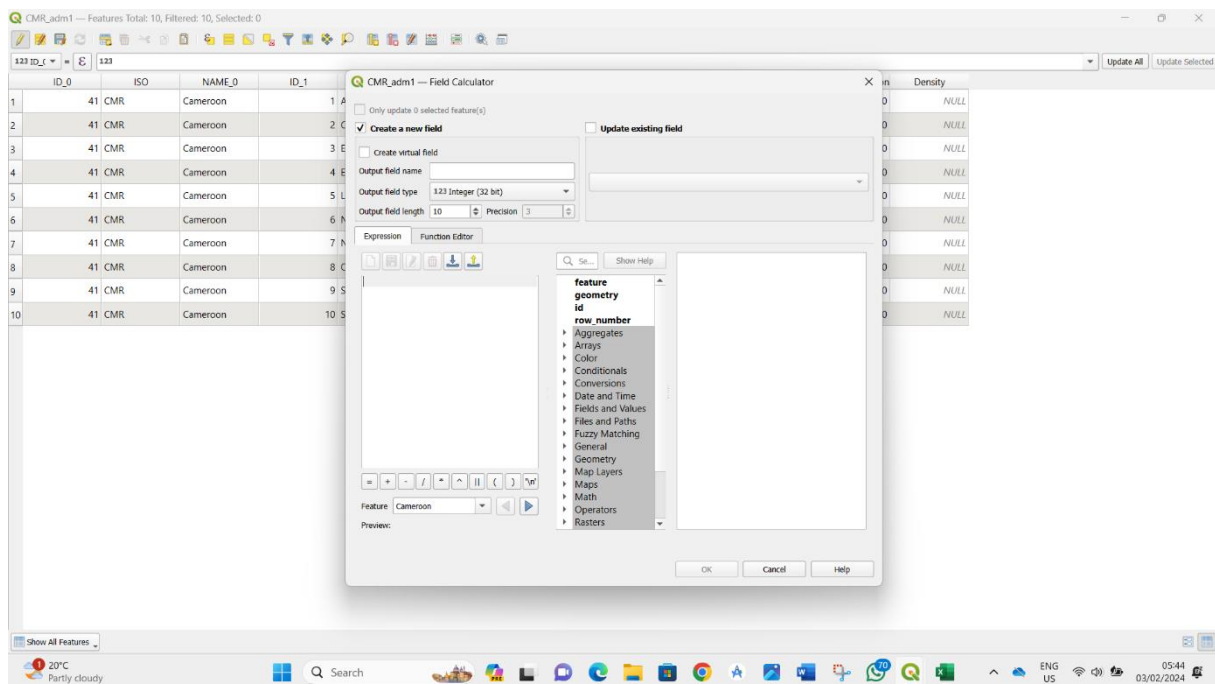
20°C
Partly cloudy

Search

ENG
US

05:41
03/02/2024

12. Then known click on your field calculator found on the menu bar in your attribute table it will display the form below:



CMR_admin1 — Features Total: 10, Filtered: 10, Selected: 0

Field Calculator

☐ Only update 0 selected feature(s)

☒ Create a new field ☐ Update existing field

Create virtual field

Output field name:

Output field type: 123 Integer (32 bit)

Output field length: 10 Precision: 3

Expression:

Function Editor

feature
geometry
id
row_number
Aggregates
Arrays
Color
Conditionals
Conversions
Date and Time
Fields and Values
Files and Paths
Fuzzy Matching
General
Geometry
Map Layers
Maps
Math
Operators
Rasters

Feature: Cameroon

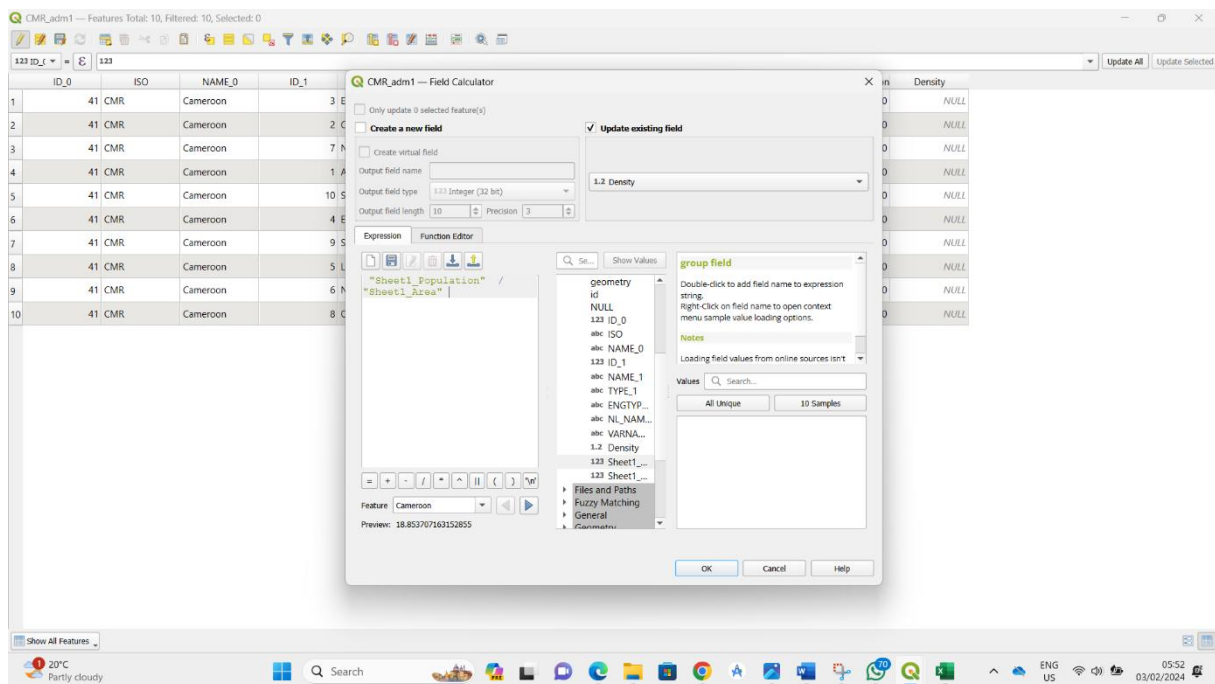
Preview:

OK Cancel Help

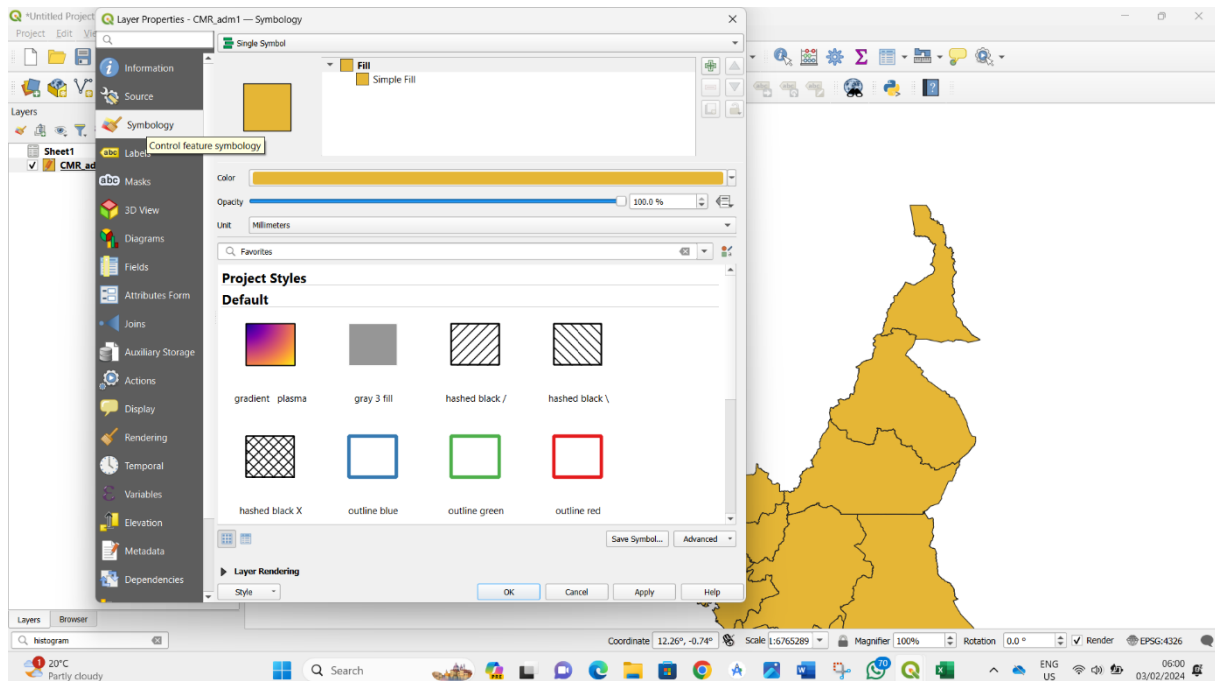
Show All Features

20°C Partly cloudy Search 05:44 03/02/2024

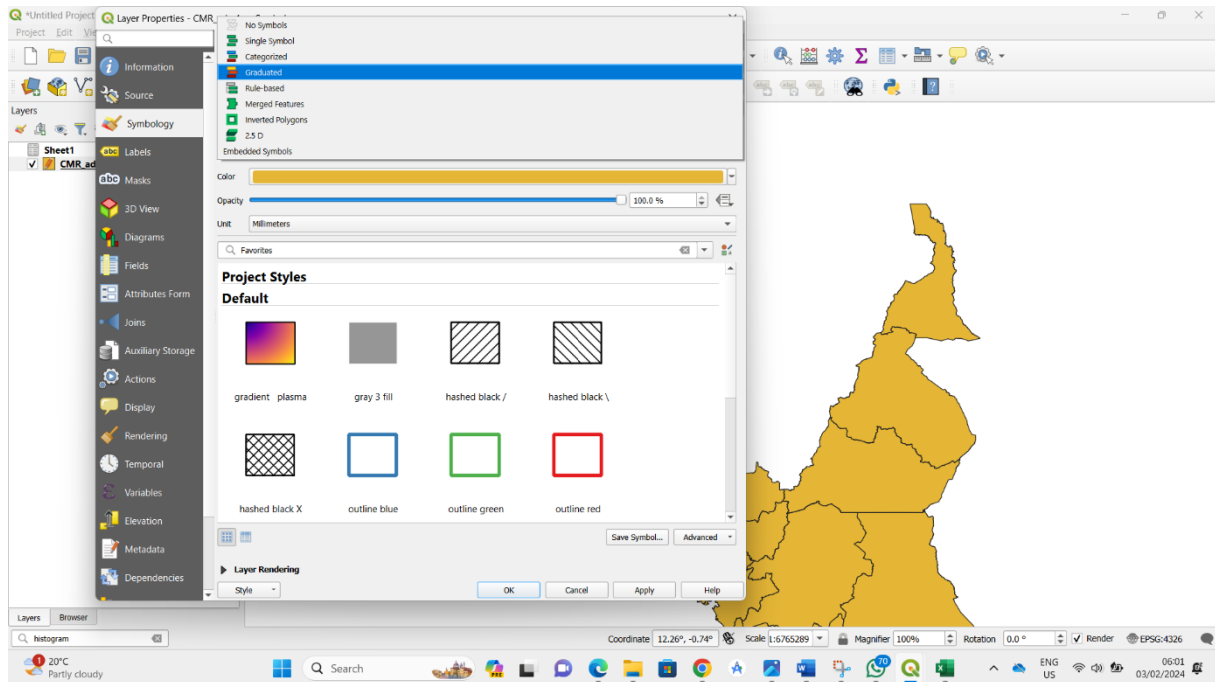
13. As shown on the form below you will tick the update existing field then below you will select our density field, and below where the list is you will select fields and values and then you will choose our population sheet(p) / surface area(s) sheet because to calculate population density is p/s which will be calculated on the expression site as shown below.



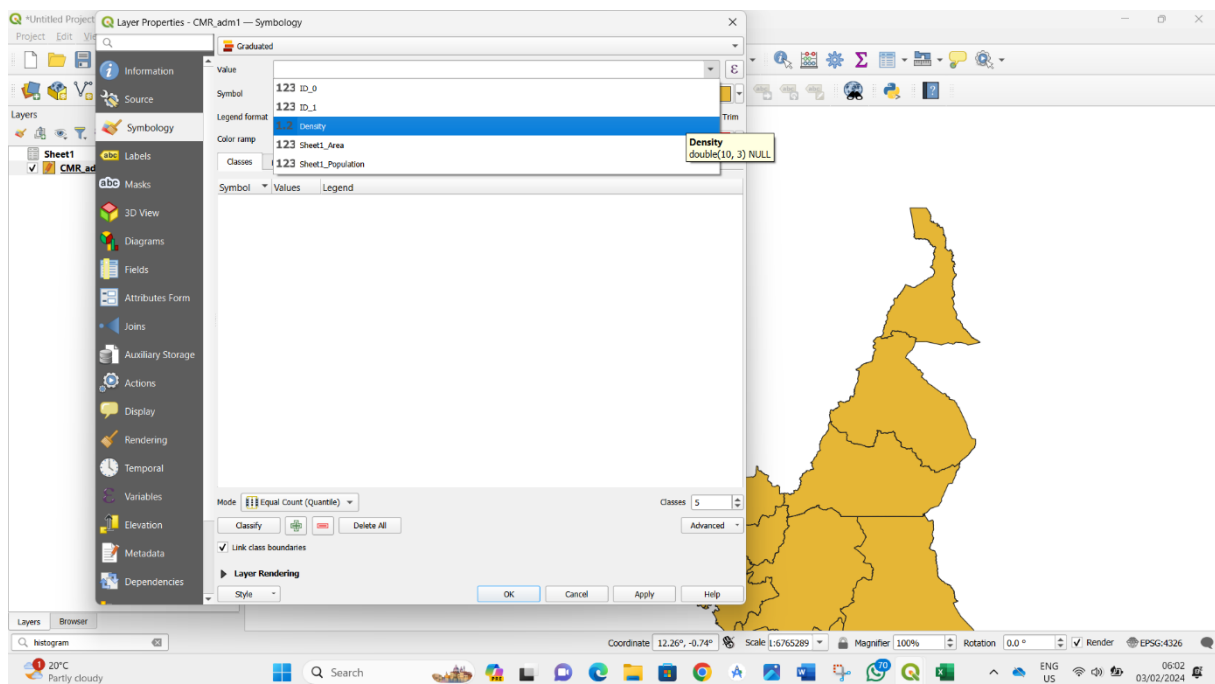
14. Then known right on your CMR_adm1 layer and go to its properties then known on the side bar choose symbology and the page below should appear on your screen:



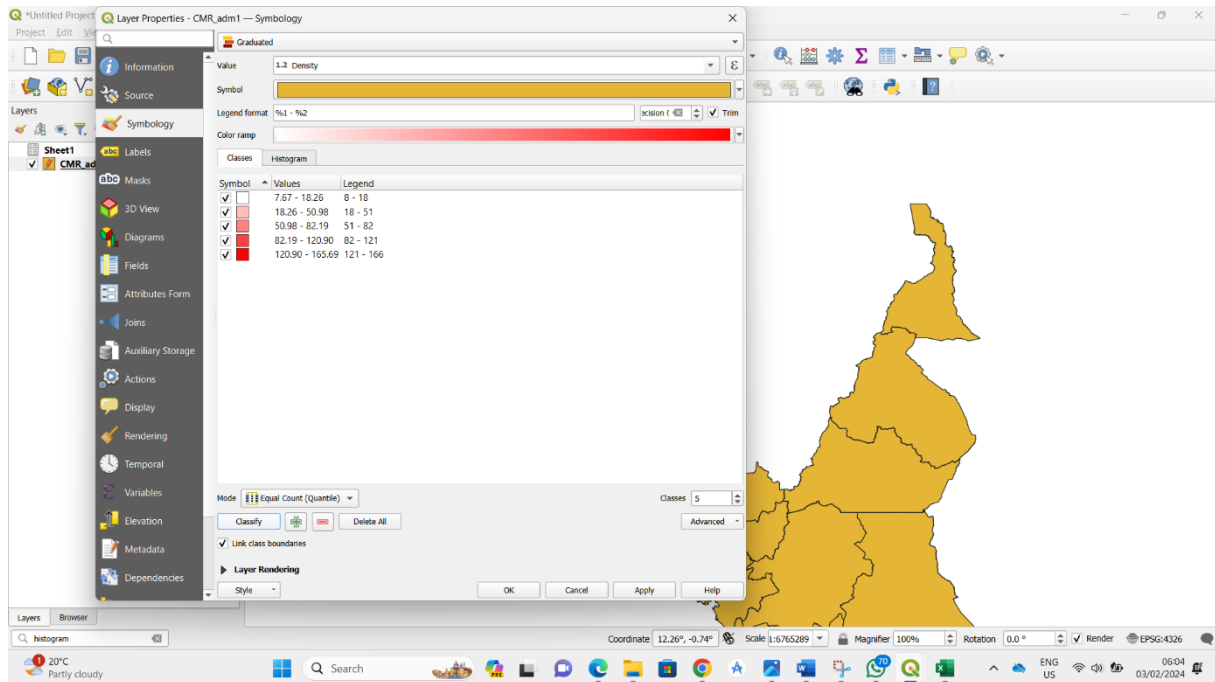
15. Known as shown below click on the drop down button of the label named single symbol and choose graduated as shown below:



16. After that at the area of symbol choose the Density column that you calculated which is found in your attribute table as shown below:

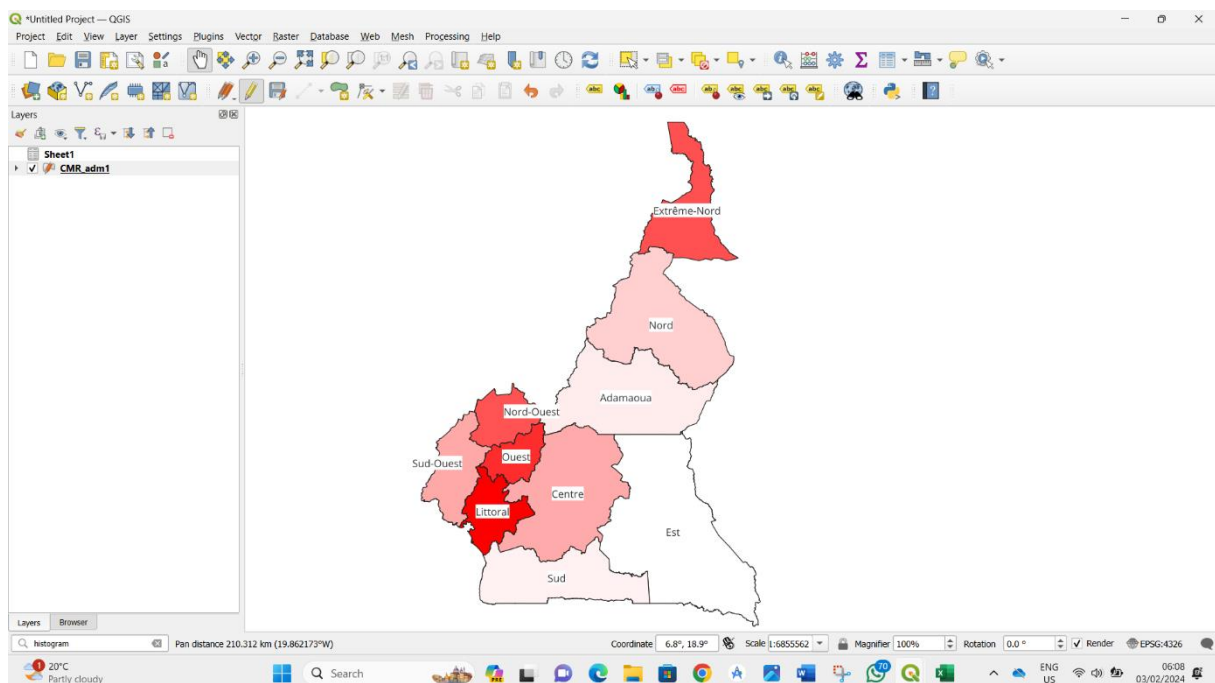


17. Then as shown below click on the classify button down for it to display the way the different population densities will be shared amongst the 10 regions of Cameroon :



Then when your done you can click on apply then you click on OK for the changes to be applied

Here is the OUTPUT OF THE population density of Cameroon distributed in the ten different regions from the lowest with a lighter colour to the highest with a more thicker colour:



Shape files

Shapefiles are a popular file format for storing geospatial data. They are used in many GIS applications, including QGIS.

A shapefile consists of several files that store different types of information, such as the geometry of the features, the attributes of the features, and the projection information. In QGIS, you can create a new shapefile by going to Layer >> Create Layer >> New Shapefile Layer.

You can also open an existing shapefile by clicking Layer >> Add Layer >> Add Vector Layer and selecting the shapefile you want to open. Once you have a shapefile loaded in QGIS, you can use it to create maps, perform spatial analysis, and more.

III. RESULTS

In this section, we present the results of our study on QGIS. We used a combination of data sources, including shape files map of land use in the study area. We employed a range of tools and techniques in QGIS to analyze the data, including spatial analysis, interpolation, and clustering. Our findings reveal several interesting patterns in land use, including a significant increase in urbanization over the past decade. We also identified several areas where land use is changing rapidly, which could have important implications for conservation efforts. Overall, our study demonstrates the power of QGIS as a tool for analyzing and visualizing geospatial data.

a. Demonstration

This will be shown on the Projector using the software QGIS3.34.

IV. DISCUSSION

Our study demonstrates the potential of QGIS as a powerful tool for analyzing and visualizing geospatial data. Our findings reveal several interesting patterns in land use, including a significant increase in urbanization over the past decade. This trend has important implications for conservation efforts, as it suggests that natural habitats are being lost at an alarming rate.

Our study also identified several areas where land use is changing rapidly, which could have important implications for urban planning and development.

For example, our analysis revealed the patterns of the roads found in Cameroon ,the different water bodies, the unexploited areas and so on . This trend could have significant environmental and social impacts, and it highlights the need for more sustainable land use practices. Overall, our study provides valuable insights into the potential of QGIS as a tool for analyzing and visualizing geospatial data, and it highlights the importance of continued research in this area.

a. **Advantages of using QGIS:**

- **Multiple language support:** QGIS is available in several languages, including English , Italian, Spanish, and French ¹.
- **Free and open-source:** QGIS is free to download and use, and its source code is available for anyone to modify and distribute ¹.
- **Cross-platform compatibility:** QGIS is compatible with Windows, Linux, and macOS operating systems ¹.
- **User-friendly interface:** QGIS has a user-friendly interface that makes it easy for new users to operate ¹.
- **Versatile:** QGIS offers a wide range of tools for spatial processing, analysis, and visualization ¹.
- **Community support:** QGIS has a large and active community of users who provide support, write code, and create new tools ¹.

In summary, QGIS is a powerful and versatile tool that is free to use and has a supportive community. It is an excellent choice for anyone who needs to work with geospatial data.

b. Disadvantages of using QGIS:

- **Steep learning curve:** QGIS has a lot of features and can be overwhelming for new users. It may take some time to learn how to use the software effectively ¹.
- **Limited support:** While QGIS has a large and active community, it may not have the same level of support as commercial GIS software ¹.
- **Limited documentation:** The documentation for QGIS can be sparse and difficult to navigate ¹.
- **Performance issues:** QGIS may be slower than commercial GIS software when working with large datasets ¹.
- **Compatibility issues:** QGIS may have compatibility issues with some file formats and data sources

V. CONCLUSION:

QGIS is a powerful and versatile tool that is free to use and has a supportive community. It is an excellent choice for anyone who needs to work with geospatial data. While QGIS has some disadvantages such as a steep learning curve, limited support, and limited documentation, it remains a popular choice for many users due to its versatility, user-friendly interface, and active community support

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qgis.org

Developer(s)QGIS Development Team

Initial releaseJuly 2002

Stable release3.34.3 / 19 January 2024

Repositorygithub.com/qgis/QGIS

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